

Low-effort determination of heat capacity and thermal conductivity for cylindrical 18650 and 21700 lithium-ion cells

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ABSTRACT

High-fidelity simulations of lithium-ion cells describing fast charging or safety-critical events often rely on precise thermal parameters. Usually, the determination of cell parameters such as heat capacity or thermal conductivity is time consuming and cost intensive. To streamline this process, a simple yet precise measurement method for determining the specific heat capacity of lithium-ion cells is now extended by an approach that also captures their thermal conductivity. The method presented here was developed using two cylindrical cell formats, each containing a lithium–nickel–manganese–cobalt-oxide (NMC) cathode and a silicon-graphite anode. The extension is based on a two-dimensional finite element (FEM) model, whose geometry is parameterized by computed tomography (CT) scans. To ensure the simplicity of the experimental evaluation, the FEM model is approximated by a rational function. The new measurement method is applied at three different temperatures and states of charge (SOC). The results show that while the specific heat capacity of the jelly roll increases with temperature by 0.2%/K for both cell formats, it is only minimally influenced by the SOC. In contrast, the through-plane thermal conductivity of the electrode-separator-composite decreases by -0.5%/K and is enhanced by up to 9% by increasing the SOC in a window ranging from 10% to 71%.

1. Introduction

Lithium-ion cells are regarded as a key element of greenhouse gas reduction in the transport sector. Despite this, short driving ranges and insufficient fast charging capabilities of battery electric vehicles continue to prevent their broad consumer acceptance. Increasing driving range by improving the energy density of the battery system often comes with a reduced share of inactive parts such as the current collectors or the electrolyte inside the cell [1]. This increase in a lithium-ion cell's energy density is accompanied by a compromised thermal performance [2], as the heat rate per volume increases whilst the heat transport to the surrounding thermal management system is diminished. Deteriorated thermal performance can reduce the fast charging capabilities as a certain temperature-threshold has to be adhered to avoid accelerated aging or safety issues [1]. For the thorough design and operation of a battery, the precise characterization of thermal parameters such as heat capacity and thermal conductivity are therefore more significant than ever.

The thermal characterization of lithium-ion batteries is time-consuming and frequently requires special equipment [3]. In addition, some techniques require the cell to be fitted with internal temperature

sensors [4], a process that increases both the level of complexity and the uncertainties. Bryden et al. [5] introduced a novel, simple, and precise measurement technique for determining the thermal heat capacity, referred to in this work as the Bryden method. This technique requires only standard laboratory equipment based around a battery test system and a climate chamber with an adjustable ventilation speed. By heating the battery cells with two different convective boundary conditions, Bryden et al. [5] were able to characterize the heat capacity of two cylindrical, one pouch, and one prismatic lithium-ion cell. They validated their approach with calorimetry measurements and reported a maximum deviation of 4% between measurements.

The Bryden method [5] determines the lumped parameters of the thermal equivalent circuit model in Fig. 1. Accordingly, the internal thermal resistance R_{in} can also be characterized by this method. An example of this is given by Akbarzadeh et al. [6]. They applied the same single capacitance network to a prismatic cell with a nominal capacity of 43 Ah. As pointed out by Fleckenstein et al. [7], evaluating the thermal conductivity using a single-capacitance network can lead to an error of over 100%. Therefore, it is questionable whether the

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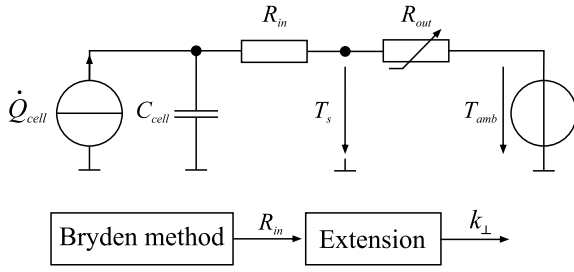


Fig. 1. Thermal equivalent circuit model applied by the Bryden method [5], which includes the heat rate inside the cell $\dot{Q}_{cell}$, the lumped heat capacity C_{cell} , the internal R_{in} and external R_{out} thermal resistance, and the ambient temperature T_{amb} of the cell. The magnitude of R_{out} can be controlled by the surrounding air velocity, which affects the temperature at the shell of the cell T_s . The extension of the Bryden method [5] is visualized by the block diagram below the circuit model.

approach of Akbarzadeh et al. [6] provides sufficient accuracy under different test conditions.

Al-Zareer et al. [8] extended the Bryden method [5] to three cylindrical 18650 cells. In contrast to Akbarzadeh et al. [6], they used a three-dimensional FEM model to evaluate the heat capacity and anisotropic thermal conductivity, thus avoiding the errors inherent in the single-capacitance network. In addition, a single-particle electrochemical model was employed to determine the heat generation inside the cell. Despite the accuracy of these methods, the FEM model introduces additional complexity that may prevent their wide application by experimental researchers.

The main objective of this study is to provide an easy yet precise extension of the Bryden method [5] that overcomes the aforementioned issues of validity and complexity. Specifically, the extension comprises a transformation of the internal resistance R_{in} into the effective thermal conductivity $k_{\perp}$, as highlighted in the block diagram in Fig. 1. Special attention is paid to a simple evaluation of the measurement data that will be demonstrated for a 18650 and a 21700 cell at different SOC and temperatures. The aim of this investigation is to supply further insights into the dependencies of thermal conductivity on temperature and SOC, as there is still no coherent agreement in the literature [3].

This paper consists of three sections. Section 2, Experimental, introduces the test setup and the cells under investigation. It also discusses the test protocol and calculates the measurement uncertainties. The evaluation of the thermal conductivity is described in Section 3. This section also contains the estimates of thermal conductivity and heat capacity; both of these thermal properties are estimated on the basis of material measurements found in the literature. Finally, Section 4 presents the results regarding the thermal conductivities and heat capacities of the 18650 and 21700 cells at different temperatures and SOC.

2. Experimental

2.1. Test setup and cells

Two cylindrical high-energy lithium-ion cells are investigated in this paper. Both cells are manufactured by LG Chem Ltd. and differ in their dimensions. According to Sturm et al. [9], a silicon-graphite anode and an NMC-811 cathode are used in both cells. The cell specifications by LG Chem Ltd. are listed in Table 1.

To apply the Bryden method [5], the battery cell has to be heated using an alternating current profile. At least two heating phases are required in the presence of different convective boundary conditions. In this study, a wind tunnel for battery cells, which is a more evolved variant of the test setup introduced by Rheinfeld et al. [10], was deployed to maintain the required boundary conditions. A schematic representation of the entire test setup is given in Fig. 2(a) and a

Table 1

Specifications of investigated cell formats according to the product specifications.

	18650	21700
Identification	INR18650 MJ1	INR21700 M50T
Manufacturer	LG Chem Ltd.	LG Chem Ltd.
Dimensions in mm	Ø18.5; 65.1	Ø21.1; 70.2
Nominal voltage	3.64 V	3.63 V
Capacity	3.35 Ah	4.85 Ah
Mass	49 g	69 g
Voltage range	[2.5; 4.2] V	[2.5; 4.2] V
Current range (continuous)	[−10; 3.35] A	[−7.28; 3.40] A

photograph is shown in Fig. 2(c). The inner air within the test setup is accelerated by two axial fans inside the left and right wing of the wind tunnel. Two heat exchangers are connected to a refrigerated bath circulator (Huber, CC-508) that controls the air temperature. The two turbulent air streams are converted to a laminar stream by an air rectifier (Wacosystems, WaveCore PET150) inside the middle tunnel. The air velocity v after the rectifier is measured by an anemometer (Höntzsch GmbH, ZS25) before it passes by the cell holder.

Fig. 2(b) shows a computer-aided design (CAD) drawing of the cell holder, to which two cells can be connected. This fixes the position of the cell inside the air stream and provides an electrical connection to the battery test system (BaSyTec, CTS). The test system controls the cell current I_{cell} and measures the cell voltage U_{cell} with four-terminal sensing. The electrical cell connection can be seen in Fig. 2(d). At Tag 2 (indicated), the sense and load connectors of the cell are soldered to a HILUMIN® steel strip that is spot welded to the cell terminal. U_{cell} therefore includes the voltage drop due to the electrical resistance between Tag 1 and 2. The magnitude of this external resistance Z_e was measured with a resistance meter (HIOKI, BT3562) and the results are listed in Table 3 with an uncertainty of 0.5%. The internal resistance Z_i of the negative and positive cell terminals were determined in the same manner after opening the cells and removing the jelly roll from the inner tab. A photograph of the test setup for the positive cell terminal with the soldered measuring wires can be seen in Fig. 2(f).

All temperatures are measured by 100 Ω platinum-chip sensors (JUMO PCA 1.2005.1L F0,15) connected to the National Instruments NI-9216 module in four-wire high-resolution mode, with a sample rate of 2 Hz. Before the sensors were attached, the Callendar-Vandusen coefficients were calibrated between -10 and 40 $^{\circ}\text{C}$ using a Fluke Calibration 1524 Thermometer and a FLUKE-TEMP-5615-6-P temperature sensor with an accuracy of 0.012 $^{\circ}\text{C}$. As can be seen in Fig. 2(b), three temperature sensors were attached to each cell, one to the shell of the cell T_s and one each to the positive T_{ip} and negative T_{in} terminal. Each sensor was glued to the metal surface and insulated from the ambient air temperature T_{amb} by a 1-cm-thick piece of insulation material (ArmaFlex®). A fully connected cell is shown in Fig. 2(e).

2.2. Test protocol and evaluation

Fig. 3 shows the overall test protocol represented by the cell voltage of the 18650 cell. The total test time of 60 days comprised three cycle tests, which were used to determine the state of health. Between the cycles, the cells were charged or discharged to a defined open circuit voltage (OCV) U_{ocv} prior to the thermal characterization. All cycle tests and SOC adjustments were conducted at an ambient temperature of 298 K. For the thermal characterization, T_{amb} was varied between 283 and 313 K.

During the cycle tests, the battery tester charged both cells at a constant current (CC) of 1 C, followed by a constant voltage (CV) step at 4.2 V. Once the cut-off current of 0.1 C was reached, the cell was left to rest for 0.75 h. Subsequently, a CC–CV discharge of 1 C to 2.5 V was conducted to determine the capacity of the cell. Both steps were repeated ten times for each cycle test. Dividing the discharged capacity during the cycle tests by the nominal capacity of the cell

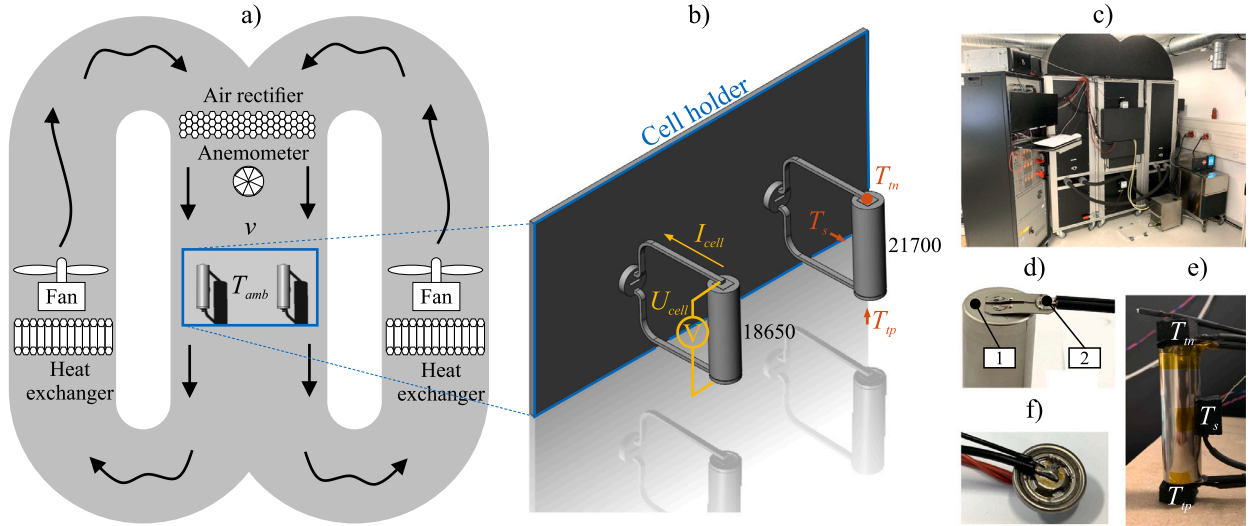


Fig. 2. Test setup for the heat capacity and thermal conductivity experiments. (a) Schematic illustration of the wind tunnel for battery cells with controllable air velocity v and ambient temperature T_{amb} . (b) Cell holder for two cylindrical cells inside the wind tunnel. (c) Photograph of the wind tunnel. (d) Electrical cell connection with measurement points for the electrical resistance Z_e at Tags 1 and 2. (e) Photograph of the attached temperature sensors, which are insulated from the ambient air. (f) Positive terminal of the 18650 cell with soldered sense and load wires for determining the internal electrical resistance Z_e .

Table 2

Relation between U_{ocv} and the resulting SOC at the end of the SOC adjustment shown in Fig. 3.

$\frac{U_{ocv}}{V}$	SOC 18650	SOC 21700
3.42	10%	15%
3.63	33%	34%
3.93	70%	71%

computes to an average state of health of 92% and 96% for the 18650 and 21700 cells, respectively. The SOC adjustments in Fig. 3 were performed by conducting CC steps of 0.05 C until a defined U_{ocv} was reached, followed by a rest period of 24 h. As can be seen in Fig. 3, an open-circuit voltage U_{ocv} of 3.42, 3.64 and 3.93 V were used to terminate the CC phase of the SOC adjustments. Using the capacity from the cycle tests and integrating the cell current I_{cell} during the SOC adjustments, a specific SOC can be assigned to each U_{ocv} . Table 2 shows the assignments between U_{ocv} and SOC for both cell formats.

For each thermal characterization step, the cell has to be heated at least twice using an alternating current profile. Fig. 4(a) shows the heating process, which is subdivided into a rest, a heating, and a cooling phase at two different air velocities v . Additionally, the temperatures at the cell terminals (T_{nt} , T_{pt}) are shown for the lower air velocity of 0.6 m s^{-1} . In Section 3.4, the increase in T_{nt} and T_{pt} is used to quantify the heat losses through the electrical connection. A sinusoidal current with a frequency of 0.05 Hz was chosen for the heating phase, as can be seen in Fig. 4(b). The amplitudes $\hat{I}_{cell}$ were set to 3.35 A and 3.5 A for the 18650 and 21700 cell, respectively. Fig. 4(b) also shows the total heat rate inside the cell $\dot{Q}_{cell}$. This is calculated using Eq. (1) by subtracting the heat loss inside the external impedance of the positive and negative cell connection Z_e from the total active power. The active power is the product of $\hat{I}_{cell}$, the amplitude of the cell voltage $\hat{U}_{cell}$ and the phase angle φ of the U_{cell} with respect to I_{cell} .

$$\dot{Q}_{cell} = 0.5 \hat{U}_{cell} \hat{I}_{cell} \cos(\varphi) - 0.5 \hat{I}_{cell}^2 \text{Re}\{Z_e\} \quad (1)$$

Conveniently, the heat rate inside the cell $\dot{Q}_{cell}$ depends only on electrical measurement quantities, which favors a small measurement uncertainty of $\dot{Q}_{cell}$. Reversible heat fractions even out along one cycle duration of the sinusoidal current [11]. Therefore, only irreversible heat contributes to the exponential temperature increase applied in the Bryden method [5]. Eq. (1) determines the resistive heat, which

Table 3

Electrical line resistances inside and outside the cell excluding the jelly roll. Z_e is the ohmic resistance between the measurement points in Fig. 2(d) for the negative and positive terminal. Z_i represents the ohmic resistance between the outer terminal and the internal connection point of the jelly roll in Fig. 2(f).

	18650	21700
External Z_e	1.03 mΩ	0.85 mΩ
Internal Z_i	2.71 mΩ	2.42 mΩ

is considered the only significant heat phenomenon, if the cell is operated with a sinusoidal current [12]. Nevertheless, the uncertainty in $\dot{Q}_{cell}$ increases if the heat produced from mixing, heat produced from phase-changes or parasitic reactions occur inside the cell.

The Bryden method [5] uses Eqs. (2) and (3) to calculate the internal thermal resistance R_{in} and heat capacity C_{cell} of the cell. The external thermal resistance R_{out} and the temperature-time constant τ can be determined for every heating procedure. For a single heating procedure, Eq. (3) contains two unknown variables R_{in} and C_{cell} . For two heating procedures with differing convective boundary conditions (τ_1 , τ_2 , R_{out1} , R_{out2}), there is a system of equations with a single solution for R_{in} and C_{cell} . The solution is given by Eqs. (4) and (5).

$$R_{out} = \Delta T_s / \dot{Q}_{cell} \quad (2)$$

$$\tau = C_{cell} (R_{in} + R_{out}) \quad (3)$$

$$R_{in} = \frac{R_{out1} - (\tau_1 / \tau_2) R_{out2}}{(\tau_1 / \tau_2) - 1} \quad (4)$$

$$C_{cell} = \frac{\tau_1 - \tau_2}{R_{out1} - R_{out2}} \quad (5)$$

In contrast to the work of Bryden et al. [5], the temperature-time constant τ in Fig. 4(a) was obtained during the cooling phase rather than the heating phase. As can be seen in Fig. 4(b), $\dot{Q}_{cell}$ is not an idealized step function. Rather, the heat rate at the beginning of the heating phase is slightly higher than at the end, causing a faster increase in T_s . This effect is very likely caused by a decrease in the electrical impedance of the cell resulting from the increasing temperature. As a result, fitting τ at the beginning of the heating phase yields approximately 25% greater values. As the cooling phase is closer to an idealized step function, τ is fitted here using the objective function $p_1 \exp(-t/\tau) + p_3$. The temperature increase (ΔT_{s1} , ΔT_{s2}) and

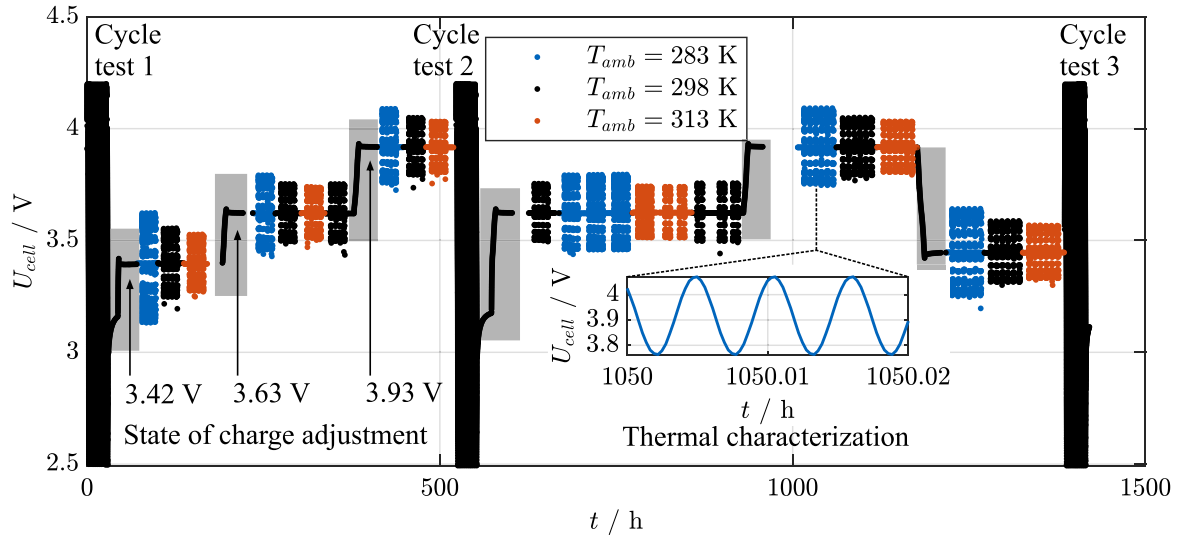


Fig. 3. Test protocol for the thermal characterization illustrated by the cell voltage of the 18650 cell. Three cycle tests determine the state of health of the cell. The thermal characterization was conducted at three different ambient temperatures and open circuit voltages of 3.42, 3.63 and 3.93 V.

Table 4

Measurement results for the lumped parameters heat capacity C_{cell} and internal resistance R_{in} for the 18650 cell at three different ambient temperatures and SOCs. For each result, the arithmetic mean and the 95% interval are given.

$\frac{U_{ocv}}{V}$	T_{amb} K	$\frac{R_{in}}{K W^{-1}}$	$\frac{C_{cell}}{J K^{-1}}$	$\frac{R_{out1}}{K W^{-1}}$	$\frac{R_{out2}}{K W^{-1}}$	$\frac{\tau_1}{s}$	$\frac{\tau_2}{s}$
3.42	283	0.373 ± 0.03	42.4 ± 0.12	13.3 ± 0.03	4.2 ± 0.00	579.6 ± 1.07	193.2 ± 0.57
	298	0.407 ± 0.02	43.5 ± 0.09	14.0 ± 0.04	4.2 ± 0.01	625.6 ± 1.53	201.8 ± 0.66
	313	0.467 ± 0.02	44.7 ± 0.09	14.3 ± 0.03	4.2 ± 0.01	661.9 ± 1.14	210.8 ± 0.47
3.63	283	0.363 ± 0.03	42.2 ± 0.16	13.3 ± 0.25	4.2 ± 0.04	577.3 ± 9.99	190.9 ± 1.74
	298	0.391 ± 0.03	43.5 ± 0.09	13.9 ± 0.11	4.2 ± 0.02	621.4 ± 3.95	199.0 ± 0.92
	313	0.424 ± 0.05	44.9 ± 0.49	14.3 ± 0.14	4.2 ± 0.03	661.0 ± 7.13	209.2 ± 2.17
3.93	283	0.347 ± 0.02	42.2 ± 0.09	13.2 ± 0.03	4.1 ± 0.00	570.0 ± 0.81	189.3 ± 0.65
	298	0.407 ± 0.02	43.4 ± 0.08	14.1 ± 0.06	4.3 ± 0.03	629.4 ± 2.43	203.0 ± 1.10
	313	0.415 ± 0.02	44.8 ± 0.06	14.4 ± 0.03	4.3 ± 0.01	664.3 ± 1.49	209.7 ± 1.06

heat generation ($\dot{Q}_{cell1}$, $\dot{Q}_{cell2}$) are taken from the measurement data at the end of the heating phase.

The experimental data in Fig. 3 were evaluated using the aforementioned procedure. The results of this evaluation are listed in Tables 4 and A.14 for the 18650 and 21700 cell. Each of the nine battery states defined by T_{amb} and U_{ocv} was repeated four to ten times. Table 4 lists the arithmetic mean and the 95% confidence interval of these repetitions. The internal resistance R_{in} derived with this method ranges between 0.347 and 0.467 K W⁻¹ for the 18650 cell. For the 21700 cell, lower values of between 0.302 and 0.437 K W⁻¹ result. Dividing the heat capacity of the cell C_{cell} by the respective cell mass m_{cell} results in a specific heat capacity c_{cell} of between 909 and 968 J kg⁻¹ K⁻¹. This range aligns with the results of 981 J kg⁻¹ K⁻¹ for the 26 650 NMC cell measured by Bryden et al. [5].

2.3. Measurement uncertainty

Measurement uncertainties can be quantified either by comparing the measurand with a reference material [3,4] or by analyzing the functional relation between the measurand and its underlying quantities [13]. For the first approach, it is important to select a reference material with similar characteristics, such as the test object [14]. For the thermal characterization of lithium-ion cells, quartz glass [4] can be used as an appropriate reference with a heat capacity of 745 J kg⁻¹ K⁻¹ and a thermal conductivity of 1.38 W m⁻¹ K⁻¹. Thermal characterization techniques which generate the heat inside the cell also need equal electrical parameters. To the best knowledge of the author no

reference exists with similar thermal and electrical properties to those of cylindrical lithium-ion cells. This also includes the anisotropic nature of the thermal conductivity of lithium-ion cells, which is not covered by any suitable reference material. Therefore, the uncertainties of the indirect measurands R_{in} and C_{cell} are computed with the uncertainties of underlying quantities such as $\hat{U}_{cell}$ or ΔT . For ease of use, the indirect quantities τ_1/τ_2 and $\Delta\tau = \tau_1 - \tau_2$ are introduced for subsequent calculations.

First the uncertainty in $\dot{Q}_{cell}$ from Eq. (1) is analyzed. The battery test system specifies a full-scale precision of 1 mV and 1 mA. Considering a cell current amplitude of 3.35 A, the uncertainty arising from the current measurement e_I calculates to 0.03%. For the 18650 cell, the smallest voltage amplitude $\hat{U}_{cell}$ is 109 mV, which results in an uncertainty of 0.91% for a precision of 1 mV. At the smallest voltage amplitude, an active power of 186 mW was measured, and the heat loss in Z_e of 6 mW can be computed at this operating point. Thus, $\dot{Q}_{cell}$ calculates to 180 mW with an absolute uncertainty for the active power of 1.748 mW ($e_U + e_I$) and an uncertainty for the heat loss in Z_e of 0.034 mW ($2e_I + 0.5\%$). Finally, the maximum uncertainty of $\dot{Q}_{cell}$ computes to 0.99%, which is the ratio of the total uncertainty of 1.782 mW and 180 mW.

The temperature increase ΔT in Fig. 4(a) was measured by the same temperature sensor, meaning that only gain and non-linearity errors contribute to the uncertainty in ΔT . Following the calibration process, the gain errors of the temperature sensor were below 0.02%. Changes in the laboratory temperature have a greater impact on the uncertainty of ΔT , which was analyzed for T_s over 24 h for both air velocities used in this study. Due to changes in the temperature of the laboratory, T_s can drift by 16 mK at 0.6 m s⁻¹ and 6 mK at 3.1 m s⁻¹. Dividing this drift by the smallest temperature increase of 2.578 and 0.689 K results in a velocity-dependent uncertainty for ΔT of 0.62% and 0.87%, respectively. As R_{out} arises from the quotient of ΔT and $\dot{Q}_{cell}$, the two errors accumulate to form e_{Rout} . Therefore, the overall error for e_{Rout1} and e_{Rout2} amounts to 1.61% and 1.86%.

The uncertainty of the temperature-time constant τ in Eq. (3) is affected by dynamic errors in T_s . This error appears directly after the heating stops at $t = 4$ h in Fig. 4(a). This error can be neglected by starting the exponential curve fitting 20 s after the end of heating. Furthermore, measurement noise and control errors in T_{amb} can affect τ . This fitting error is identified by the 95% confidence interval. Accordingly, τ_1 exhibits an uncertainty of 0.02% and τ_2 an uncertainty of 0.1%. This translates into an uncertainty of 0.08% for the difference $e_{\Delta\tau}$ and 0.12% for the quotient e_{τ_1/τ_2} .

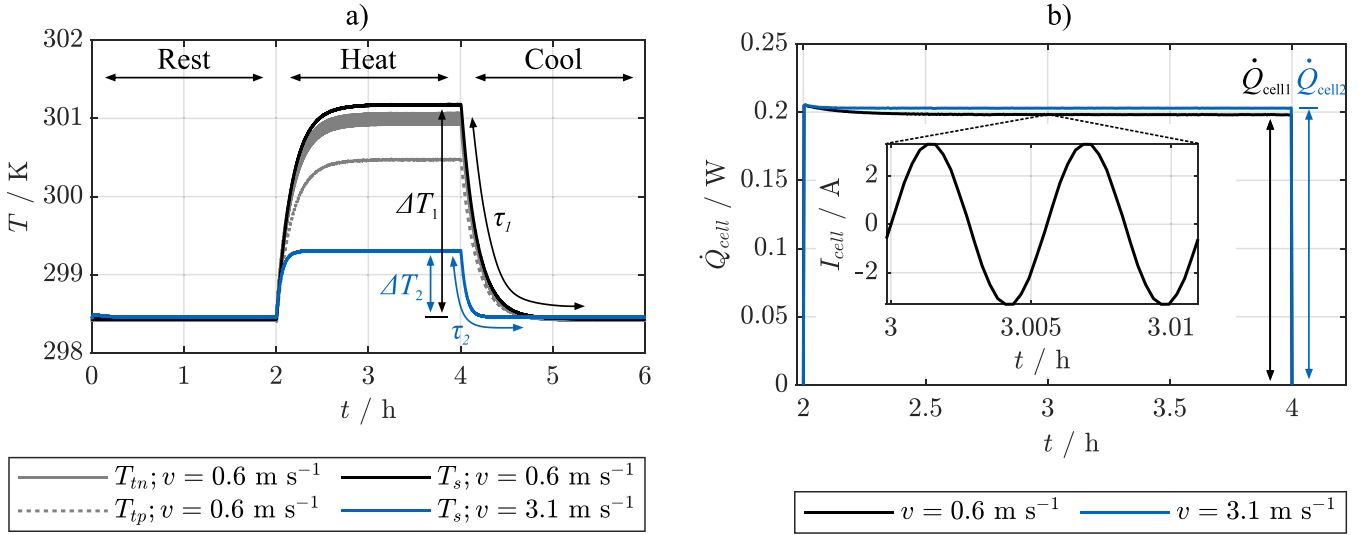


Fig. 4. Measurement readings during thermal characterization of the 18650 cell at $U_{ocv} = 3.63 \text{ V}$ with parameters for evaluating the heat capacity and thermal conductivity. (a) Cell temperature at the shell T_s during the rest, heating, and cooling phases at two different air velocities v of 0.6 and 3.1 m s^{-1} . For 0.6 m s^{-1} , the temperatures at the negative T_{tn} and positive terminal T_{tp} are displayed. Thicker lines indicate a higher alternating component arising from the sinusoidal cell current. (b) Heat rate inside the cell $\dot{Q}_{cell}$ during the heat phase caused by a sinusoidal cell current I_{cell} with an amplitude of 3.35 A and a cycle duration of 20 s.

Once the uncertainties of the measurement quantities have been analyzed, their propagation in Eqs. (4) and (5) can be computed. The relative uncertainty e_C of C_{cell} is defined by Eq. (6). Applying the linear term of the Taylor series for C_{cell} leads to Eq. (7). Eq. (8) formulates the relation between the uncertainty e_C and the measurement uncertainties e_i with their gain factors f_i . The gain factors prior to the difference of $e_{R_{out1}}$ and $e_{R_{out2}}$ is 0.230 for $R_{out1} = 14.4$ and $R_{out2} = 4.3$ (values are taken from Table 4). To determine the overall uncertainty for C_{cell} , the root sum square $\sqrt{\sum (f_i e_i)^2}$ of the measurement uncertainties e_i and their gain factors f_i is computed, which aligns with the procedure used by Sheng et al. [13]. Accordingly, the uncertainty e_C amounts to $\pm 1.62\%$, which is the same order of magnitude as the deviation from calorimetry measurements of up to $\pm 3.9\%$ measured by Bryden et al. [5].

$$e_C = \frac{C_{cell} - \text{Taylor}(C_{cell}; \Delta\tau, R_{out1}, R_{out2})}{C_{cell}} \quad (6)$$

$$e_C = \frac{\frac{\partial C_{cell}}{\partial \Delta\tau} \Delta\tau e_{\Delta\tau} + \frac{\partial C_{cell}}{\partial R_{out1}} R_{out1} e_{R_{out1}} + \frac{\partial C_{cell}}{\partial R_{out2}} R_{out2} e_{R_{out2}}}{-C_{cell}} \quad (7)$$

$$e_C = \sum (f_i e_i) = -e_{\Delta\tau} + e_{R_{out1}} + \frac{-R_{out2}}{R_{out1} - R_{out2}} (e_{R_{out2}} - e_{R_{out1}}) \quad (8)$$

The uncertainty e_R for R_{in} is defined analogously to e_C using Eq. (9). Like Eqs. (7), (10) uses the Taylor series to separate different uncertainties. In the case of R_{in} , the quotient of τ_1 and τ_2 is used instead of $e_{\Delta\tau}$. Eq. (11) shows how e_R is affected by uncertainties in e_{τ_1/τ_2} and R_{out} . According to the three gain factors f_i , the uncertainty is minimized if the difference between R_{out1} and R_{out2} is maximized and the average of both thermal resistances is as low as possible. Using the values for R_{out} and τ in Table 4 at $U_{ocv} = 3.93 \text{ V}$ and $T_{amb} = 313 \text{ K}$ the gain factors f_i in Eq. (11) are 19.0 for e_{τ_1/τ_2} , 1.0 for $e_{R_{out1}}$, and 17.5 for the difference in $e_{R_{out2}}$. This highlights the importance of low uncertainties for e_{τ_1/τ_2} and $(e_{R_{out2}} - e_{R_{out1}})$. Finally, e_R can be determined by the root sum square, which amounts to an uncertainty of $\pm 5.19\%$.

$$e_R = \frac{R_{in} - \text{Taylor}(R_{in}; \tau_1/\tau_2, R_{out1}, R_{out2})}{R_{in}} \quad (9)$$

$$e_R = \frac{\frac{\partial R_{in}}{\partial \frac{\tau_1}{\tau_2}} \frac{\tau_1}{\tau_2} e_{\tau_1/\tau_2} + \frac{\partial R_{in}}{\partial R_{out1}} R_{out1} e_{R_{out1}} + \frac{\partial R_{in}}{\partial R_{out2}} R_{out2} e_{R_{out2}}}{-R_{in}} \quad (10)$$

$$e_R = \sum (f_i e_i) = \frac{R_{out2} - R_{out1}}{\frac{\tau_1}{\tau_2} R_{out2} - R_{out1}} \frac{\frac{\tau_1}{\tau_2}}{\frac{\tau_1}{\tau_2} - 1} e_{\tau_1/\tau_2} - e_{R_{out1}} + \frac{-\frac{\tau_1}{\tau_2} R_{out2}}{\frac{\tau_1}{\tau_2} R_{out2} - R_{out1}} (e_{R_{out2}} - e_{R_{out1}}) \quad (11)$$

In conclusion, the maximum uncertainty e_C for determining C_{cell} using the Bryden method [5] amounts to $\pm 1.62\%$ in this paper. For R_{in} , the uncertainty is $\pm 5.19\%$, which is three times the value of e_C . This illustrates the necessity of precise measurement equipment for determining R_{in} . Furthermore, the thermal through-plane conductivity $k_{\perp}$ is the final quantity of interest, not R_{in} . Therefore, further evaluation steps with additional uncertainties are necessary, which will be discussed in Section 3.4.

3. Theory

3.1. Thermal cell model

Eqs. (4) and (5) of the Bryden method [5] are based on the lumped capacitance model in Fig. 1. The model assumption constitutes a simplification of the energy conservation equation, which is considered valid for Biot numbers smaller than 0.1 [15]. For cylindrical cells, the Bi can be computed by the ratio of the heat transfer coefficient h_s and the thermal conductivity $k_{\perp}$ multiplied by the cell radius [16]. As discussed in Section 2.3, a high heat transfer coefficient at the cell surface h_s has to be used for at least one measurement to ensure an acceptably low measurement uncertainty for R_{in} . This suggests the existence of a conflict between reducing measurement uncertainty and maintaining the validity of the model assumption. Dividing the inverse of the external resistance R_{out1} of 14 K W^{-1} in Table 4 by the shell area of the 18650 cell results in a value for h_s of $19.4 \text{ W m}^{-2} \text{ K}^{-1}$. This computes to a Biot number of 0.17, assuming a through-plane thermal conductivity $k_{\perp}$ of $1 \text{ W m}^{-1} \text{ K}^{-1}$ and a cell radius of 9 mm. As the Biot number exceeds 0.1 for both air velocities, a spatially resolved thermal model is used to extend the Bryden method [5].

The thermal model is created using COMSOL Multiphysics 5.4. The two-dimensional (2D) axisymmetric geometry of the 18650 cell

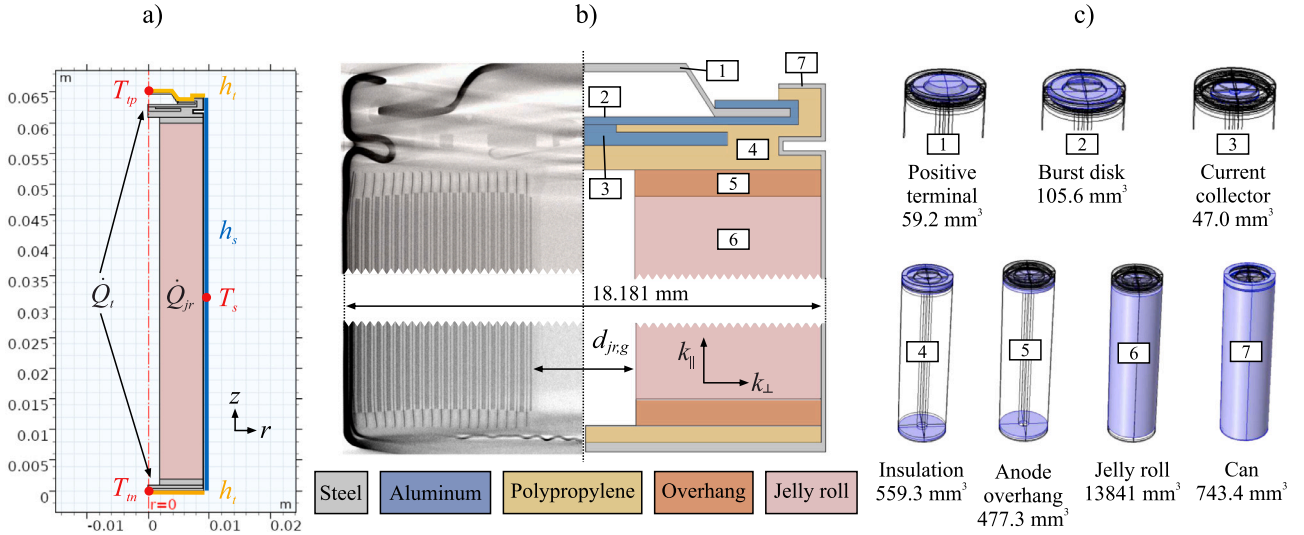


Fig. 5. Geometry and boundary conditions for the two-dimensional thermal model of the 18650 cell. (a) Thermal boundary conditions at the cell terminals h_t and the shell h_s , heat rates inside the terminals $\dot{Q}_t$, and the jelly roll $\dot{Q}_{jr}$, and positioning of the temperature sensor T_s , T_m , and T_{ip} . (b) Detailed representation of the positive and negative terminal with dimensions taken from the illustrated CT images, and the definitions of seven cell domains. (c) Three-dimensional solids of revolution of the specific cell domains including their respective volume fractions.

is depicted in Fig. 5(a). A detailed description of the model equations and parameters can be found in Tables A.18 and A.20 respectively. All dimensions were either measured with a micrometer screw or taken from CT scans, as can be seen in Fig. 5(b). Beside the CT scan, seven different domains are illustrated for the cell model. The jelly roll gap with a diameter $d_{jr,g}$ of 3.82 mm for the 18650 cell is treated as a void volume. According to Li et al. [17], the top cap of the 18650 cell contains a positive terminal (Tag 1), a burst disk (Tag 2), and a current collector (Tag 3), all of which are insulated against the steel can by a sealing ring (Tag 4). The electrode-separator composite below the top cap is divided into an active jelly roll domain (Tag 6) and an inactive anode overhang (Tag 5). Solids of revolution of each two-dimensional domain are given in Fig. 5(c) along with their volume fractions.

Inside the cell, the total heat rate $\dot{Q}_{cell}$ is subdivided into the heat rate inside the active jelly roll $\dot{Q}_{jr}$ (Fig. 5, Tag 6) and inside the cell terminals $\dot{Q}_t$. As illustrated in Fig. 5(a), different convective boundary conditions are used for the terminals h_t and the shell h_s . The terminals are insulated by ArmaFlex® material with a thickness of 10 mm and a thermal conductivity of $0.035 \text{ W m}^{-1} \text{ K}^{-1}$. Neglecting the heat capacity of this volume, h_t can be determined by dividing the thermal conductivity by the thickness ($h_t = 3.5 \text{ W m}^{-2} \text{ K}^{-1}$). However, heat conduction can also occur through the electrical connection illustrated in Fig. 2(d). Therefore, the value of $3.5 \text{ W m}^{-2} \text{ K}^{-1}$ can only be treated as a lower bound for h_t . For this reason, the coefficients h_t and h_s are determined in Section 3.4 by fitting T_s , T_{ip} , and T_m of the thermal model to the measurement data. Each material parameter in the model is considered isotropic, except for the thermal conductivity, which is defined for the in-plane $k_{||}$ and through-plane directions $k_{\perp}$ of the cell's jelly roll. Due to the homogenization of the jelly roll, $k_{\perp}$ describes the thermal conductivity of each cell layer, thermal contact resistances between the layers [18] and the jelly-roll/case interface [19], and any effect arising from the homogenization of the spiral geometry [20].

3.2. Heat capacity estimation

For the 2D thermal model of the cell, the specific heat capacity c_{jr} and the volume density ρ_{jr} of the jelly roll must be specified. Both parameters are estimated from material properties given in the literature and will be compared with experimental results. As with the anode and cathode coatings, the separator is composed of porous layers consisting of a solid and a fluid phase. If the fluid phase contains

electrolyte, it is henceforth referred to as a wet layer. If gases such as argon fill the voids, the layer is labeled dry. The volume fraction of the solid phase ε_s can be divided into an active material fraction ε_{AM} and a fraction containing the binder and carbon black (BCB) ε_{BCB} .

Loges et al. [21] defined the bulk density $\bar{\rho}$ and specific heat capacity $\bar{c}$ of the porous layers by way of Eqs. (12) and (13). The parameters used in both equations for each unit cell layer are listed in Table A.15. The results for $\bar{\rho}$ and $\bar{c}$ for each wet layer are shown in Table 5 for the 18650 cell and in Table A.17 for the 21700 cell. The mass of each layer m_i can be derived from $\bar{\rho}$, the thickness $d_{t,i}$ and length $d_{l,i}$ of the unit cell layers, and the height of the active jelly roll. The active jelly roll height is equal to the height of the Al collector for this calculation, which amounts to 58 and 63.5 mm [22] [9] for the 18650 and 21700 cell respectively. The sum of m_i in Table 5 (18650 cell) results in a total mass of the active jelly roll of 38.6 g, which is dominated by the anode and cathode coating. For the overhang mass, the overhanging height of the anode electrode and the separator of 2 mm (18650 cell) and 1.5 mm (21700 cell) are used in Table 6.

In addition to the mass of the jelly roll, the total masses of the non-jelly-roll domains m_{na} from Fig. 5(c) are listed in Table 6 computed by volumetric integration of the material densities. The total mass of the cell m_{cell} can be derived from the sum of the calculated jelly roll mass, the mass of the overhang, and the non jelly roll domains. To validate m_{cell} , the cell mass of each format was determined (Sartorius Talent TE313S, $\pm 1 \text{ mg}$) without the insulation tube, with a deviation from the calculated mass of 0.2% and 0.4% for the 18650 and the 21700 cell respectively. Furthermore, both cell formats were opened inside a glove box to extract and weigh the non-jelly-roll mass m_{na} . For m_{na} , the maximum deviation of the measured values from the calculated is 2.8%. Due to the small deviation of m_{cell} and m_{na} , the jelly roll mass and the overhang mass can also be treated as validated.

$$\bar{\rho} = \rho_F \varepsilon_F + \rho_{AM} \varepsilon_{AM} + \rho_{BCB} \varepsilon_{BCB} \quad (12)$$

$$\bar{c} = \frac{c_F \rho_F \varepsilon_F + c_S \rho_S \varepsilon_S}{\bar{\rho}} \quad (13)$$

$$c_{jr} = \frac{\sum_{i=1}^n \bar{\rho}_i d_{t,i} d_{l,i} \bar{c}_i}{\sum_{i=1}^n \bar{\rho}_i d_{t,i} d_{l,i}} \quad (14)$$

The heat capacity of the jelly roll can be derived using the weighted average in Eq. (14). The layer thicknesses $d_{t,i}$ and lengths $d_{l,i}$ are listed in Tables 5 and A.17. As the electrodes are coated onto both sides of the current collectors, the accumulated coating length is approximately

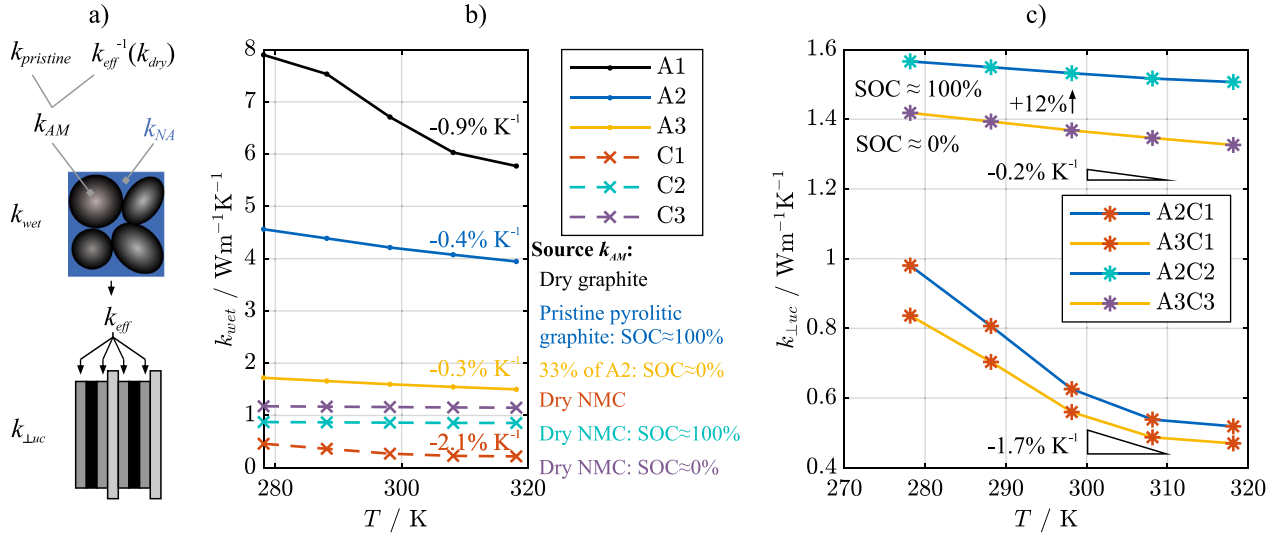


Fig. 6. Calculated thermal conductivity of the unit cell k_{LUC} based on material properties taken from the literature, which are listed in Table A.16. (a) Process of gaining the wet conductivity k_{wet} either by pristine $k_{pristine}$ or dry material k_{dry} measurements and the Bruggeman theory in Eq. (18). (b) Result for k_{wet} for different graphites (A1, A2, A3) and NMC coatings (C1, C2, C3). (c) Through-plane conductivity k_{LUC} for the 18650 cell with predicted dependencies on temperature and SOC.

Table 5

Structural parameters of wet 18650 unit cell layers and their temperature-dependent heat capacity $\bar{c}/(\text{J kg}^{-1} \text{K}^{-1}) = c_0 + c_1(T/\text{K} - 300)$.

	ϵ_F [22]	ϵ_{BCB} [22]	$d_{lt,i}$ μm [22]	$d_{ll,i}$ m [22] ^a	$\bar{\rho}_i$ g cm^{-3} ^b	m_i g ^c	c_0 ^d	c_1 ^d
Anode	0.216	0.090	86.7	1.315	2.03	13.5	881.9	2.815
Cathode	0.171	0.084	66.2	1.230	3.87	18.3	852.5	1.733
Separator	0.450	0.000	12.0	1.400	1.06	1.0	1515.5	4.573
Cu-collector	0.000	0.000	11.0	0.700	8.96	4.0	385.6	0.099
Al-collector	0.000	0.000	17.3	0.670	2.70	1.8	892.4	0.635

^aFor double-coated electrodes, the length is accumulated.

^bBulk densities of porous layer $\bar{\rho}_i$ are calculated using Eq. (12) and the material densities in Table A.15.

^cThe total layer mass m_i is derived from the product of the layer thickness $d_{lt,i}$ and length $d_{ll,i}$, bulk density $\bar{\rho}_i$ and active jelly roll height of 58 mm [22].

^dCalculated with Eq. (14).

Table 6

Weight of cell components with calculated temperature dependent heat capacities of non-active cell parts c_{na} , anode overhang c_{ao} and the jelly roll c_{jr} : $\bar{c}/(\text{J kg}^{-1} \text{K}^{-1}) = c_0 + c_1(T/\text{K} - 300)$.

	18650	21700
Jelly roll mass $m_{jr} = \sum m_i$	38.6 g ^c	56.7 g ^c
Overhang mass m_{ao}	0.6 g ^c	0.7 g ^c
Non jelly roll mass m_{na}	(7.3 ^c ; 7.1 ^m) g	(11.4 ^c ; 11.5 ^m) g
Total cell mass m_{cell}	(46.5 ^c ; 46.4 ^m) g	(68.8 ^c ; 68.5 ^m) g
Jelly roll c_{jr} : c_0	834	878.8
Jelly roll c_{jr} : c_1	1.965 K ⁻¹	1.975 K ⁻¹
Overhang c_{ao} : c_0	809.9	859.7
Overhang c_{ao} : c_1	2.325 K ⁻¹	2.299 K ⁻¹
Non jelly roll c_{na} : c_0	570.1	565.4
Non jelly roll c_{na} : c_1	0.949 K ⁻¹	0.9356 K ⁻¹

m: measured. c: calculated by volume integration with the 2D thermal model.

twice the collector length. Applying Eq. (14) with the layer parameters in Table 5 results in a temperature-dependent heat capacity of the jelly roll c_{jr} . The absolute values c_0 and temperature slope c_1 of c_{jr} are listed in Table 6 for both cells. Dividing the temperature slope of $1.965 \text{ J kg}^{-1} \text{K}^{-2}$ by c_{jr} at 300 K results in a temperature dependency of $0.2 \% \text{K}^{-1}$ for the heat capacity of the 18650 cell. Finally, we point out that an uncertainty of $\pm 8.2\%$ for c_{jr} is calculated in Table A.19.

3.3. Thermal conductivity estimation

In this section, the through-plane and in-plane thermal conductivity are calculated based on material measurements taken from the literature. This investigation allows the parametrization of various domains in the thermal model along with a more sophisticated analysis of the measurement results, which is presented in Section 4. It should be noted that this section concentrates on the thermal conductivity of the unit cell volume, which comprises two anode and cathode coatings, two separator sheets and two current collectors made of Al and Cu. Thermal contact resistances are not considered in the theoretical calculation, which is a common assumption in the literature [11]. Due to this assumption, the measured through-plane conductivity $k_{\perp}$ (Fig. 5) can differ from the calculated conductivity k_{LUC} if thermal contact resistances are significant large [15]. The discussion of $k_{\perp}$ in Section 4 will return to the neglect of thermal contact resistances if a deviation between $k_{\perp}$ and k_{LUC} is observed.

The thermal conductivity of the unit cell is defined by Eqs. (15) and (16). The thicknesses of the porous layers have to be considered twice for the unit cell. The layer thicknesses $d_{lt,i}$ are taken from Tables 5 and A.17. For an absolute estimation of k_{LUC} and $k_{\parallel}$, an isotropic thermal conductivity of 0.83 and $1.44 \text{ W m}^{-1} \text{K}^{-1}$ [18] is used for the wet NMC and graphite coatings. Thermal conductivity values for the current collectors are listed in Table A.15 and a value of $0.615 \text{ W m}^{-1} \text{K}^{-1}$ [3] for the separator is assumed. For the 18650 cell, these parameters result in through-plane k_{LUC} and in-plane $k_{\parallel}$ conductivities of 1.122 and $24.72 \text{ W m}^{-1} \text{K}^{-1}$ at room temperature, respectively. For the 21700 format, k_{LUC} and $k_{\parallel}$ calculate to 1.08 and $20.71 \text{ W m}^{-1} \text{K}^{-1}$, respectively.

$$k_{LUC} = \frac{\sum_{i=1}^n d_{lt,i}}{\sum_{i=1}^n (d_{lt,i}/k_i)} \quad (15)$$

$$k_{\parallel} = \frac{\sum_{i=1}^n (d_{lt,i} \cdot k_i)}{\sum_{i=1}^n d_{lt,i}} \quad (16)$$

Table A.19 determines an uncertainty of $\pm 21.4\%$ for k_{LUC} , which is mostly affected by the porous layers k_i in Eq. (15). On the other hand, $k_{\parallel}$ is mainly influenced by the current collectors [3], for which accurate material values for k_i are available in the literature [23] and the dependency on SOC or temperature can be neglected. $k_{\parallel}$ is therefore considered to be constant throughout this paper. In contrast, Loges et al. [24] measured a negative dependency for NMC and graphite

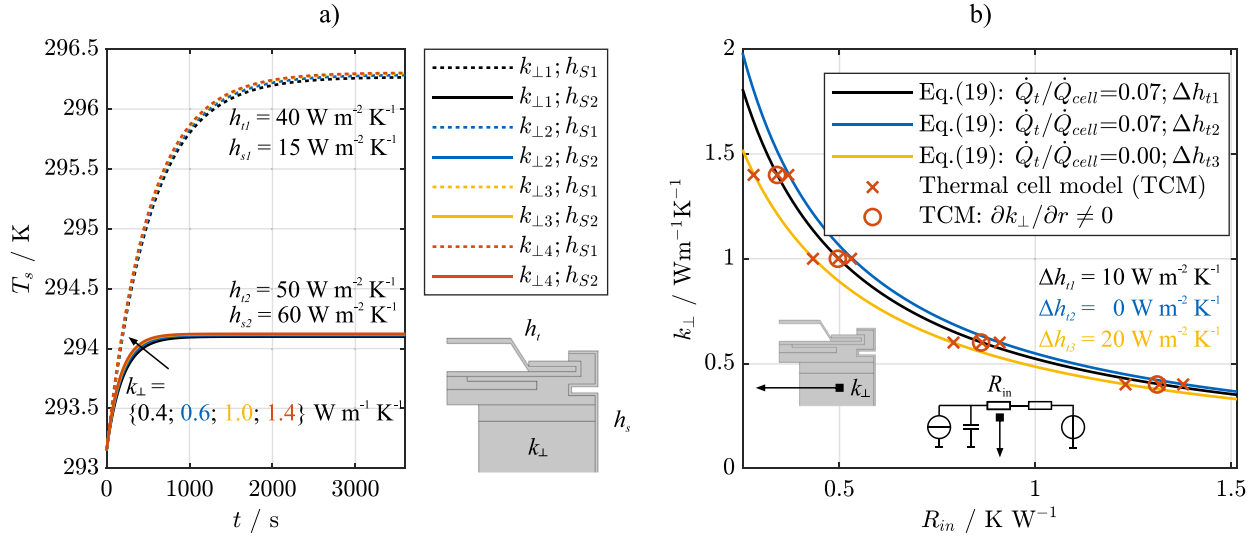


Fig. 7. Relation between lumped thermal resistance R_{in} in Eq. (4) and the through-plane thermal conductivity $k_{\perp}$ of the thermal model from Fig. 5. (a) Simulated heating phase of the thermal model for the 18650 format with varying through-plane conductivities ($k_{\perp} - k_{\perp 4}$). (b) Relation between $k_{\perp}$ and R_{in} with corresponding Eq. (19) for different heat fractions $\dot{Q}_t/\dot{Q}_{cell}$ inside the cell terminals and varying heat transfer coefficient between the heating phases Δh_{t1} . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

coatings. The effects of temperature on $k_{\perp uc}$ therefore need to be considered. Hardly any temperature-dependent measurement data for wet coatings is available in the literature. To overcome this, a method based on the work of Vadakkepatt et al. [25] for computing wet coatings from pristine materials or dry coatings is presented in the following paragraph.

The procedure for calculating the thermal conductivity of wet porous media k_{wet} is illustrated in Fig. 6(a). k_{wet} is derived with the aid of a model theory, based on the pristine thermal conductivity of non-active components k_{NA} and the active material k_{AM} . The Bruggeman model with varying exponent β developed by Vadakkepatt et al. [25] is utilized for subsequent calculations of the effective conductivity k_{eff} , which is widely applied in the lithium-ion battery community. This theory states that the thermal conductivity of the non-active material k_{NA} is calculated according to Eq. (17) [25]. The thermal conductivity of the mixture of binder and carbon black k_{BCB} is assumed to be equal to the conductivity of polyvinylidene fluoride (PVDF), listed for reference in Table A.16. For the fluid phase k_F , the thermal conductivity of LP50 electrolyte from BASF is used throughout this paper. The temperature-dependent conductivity of the electrolyte mixture LP50 is also listed in Table A.16. Please note, that the Bruggeman model only admits spherical particles, which are sufficiently far apart that their temperature field do not interact [25]. Therefore, these model assumptions are a possible source of uncertainty for the following calculations.

$$k_{NA} = \frac{\varepsilon_{BCB} k_{BCB} + \varepsilon_F k_F}{\varepsilon_{BCB} + \varepsilon_F} \quad (17)$$

$$k_{eff} = k_{AM} \varepsilon_{AM}^{\beta} + k_{NA} (1 - \varepsilon_{AM})^{1.5} \quad (18)$$

If the thermal conductivity k_{AM} and the volume fraction ε_{AM} of the active material is known, the effective thermal conductivity of the porous material k_{eff} can be estimated using Eq. (18) [25]. The Bruggeman exponent β is dependent on the ratio of k_{NA}/k_{AM} , which is linearly interpolated in this paper according to Table 7. If the k_{AM} for a certain active material is not available in the literature, it can be derived by dry measurement. In this case, the thermal conductivity of gases like argon or CO₂ has to be used for k_F and the dry measurement is considered to be k_{eff} . The thermal conductivity k_{AM} can then be calculated using the inverse function of the Bruggeman model k_{eff}^{-1} .

The results obtained by Eq. (18) for three wet anode (A1, A2, A3) and cathode (C1, C2, C3) coatings of the 18650 cell are shown in

Table 7

Relation between Bruggeman's exponent β and the thermal conductivity ratio k_{NA}/k_{AM} of non-active to active material used in Eq. (18) [25].

k_{NA}/k_{AM}	β
0.0000	2.8000
0.0027	2.5206
0.0138	2.3274
0.1025	1.6137
0.2119	1.2528
0.3169	1.0473
0.4244	0.9144
0.9982	0.9071

Fig. 6(b). The results for A1 and all cathode coatings are based on dry measurements, while those for the A2 and A3 coatings are taken from pristine data sources given in Table A.16. According to these calculations, the anode coatings range between 1 and 8 W m⁻¹ K⁻¹. This represents an overestimation in comparison with the wet measurements of graphite given in the literature, which range between 0.36 and 1.58 W m⁻¹ K⁻¹ [3]. This overestimation of k_{wet} is due to model uncertainties of the applied Bruggeman model or the low anode porosity of the 18650 cell. For example, the average porosity ε_F of the six anode coatings measured by Loges et al. [24] amounts to 50.6%. A1 is discarded from further investigation due to the implausibly high values. In contrast, the estimations for the NMC cathode coatings of 0.218 to 1.176 W m⁻¹ K⁻¹ are closer to the measured range of 0.54 to 0.91 W m⁻¹ K⁻¹ reported in the literature [3]. Finally, it should be noted that all six estimations in Fig. 6(b) exhibit decreasing thermal conductivity with increasing temperature.

Fig. 6(b) shows plots of A2 and A3 to estimate the impact of the full-cell SOC on k_{wet} . Both lines are based on pristine pyrolytic graphite [26], but A3 is reduced to 33% of A2, since, as Ju et al. [27] pointed out, that LiC₁₂ can cause such a decrease. For LiC₆, the thermal conductivity reached approximately the same value as pristine graphite. The delithiated and lithiated NMC coatings C2 and C3 measured by Gotcu et al. [28] show decreasing k_{wet} with increasing SOC. According to this analysis, graphite and NMC exhibit an opposing dependency on SOC.

Fig. 6(c) shows the through-plane conductivity $k_{\perp uc}$ for the 18650 unit cell specified in Table 5, based on the four combinations of wet coatings shown in Fig. 6(b). A constant value of 0.615 W m⁻¹ K⁻¹ [3]

for the separator layer was used in all combinations. The temperature dependency lies within -0.2 and $-1.7\% \text{ K}^{-1}$ depending on the cathode k_{wet} . As A3 and C3 can be considered as fully lithiated NMC and delithiated graphite, the SOC of 0% is assigned to the combination A3C3. The inverse situation applies for the combination A2C2, which is therefore considered the through-plane conductivity at 100% SOC. Both SOC assignments are approximations, as the degree of lithiation does not operate completely between 0.0 and 1.0 for both active materials. For example, if the SOC of the 18650 cell increases from 0% to 100%, the degree of lithiation of the anode increases from 0 to 0.85, while the degree of lithiation of the cathode decreases from 0.94 to 0.22 [22]. Based on these assumptions, the through-plane conductivity increases by 12% if the SOC increases from 0% to 100%, as highlighted in Fig. 6(c).

3.4. Thermal property evaluation

In this subsection, a functional relation is developed between the internal thermal resistance R_{in} and the through-plane conductivity $k_{\perp}$. For this relation, the simulation results of the thermal cell model in Section 3.1 are used instead of experimental data. With this technique, $k_{\perp}$ is a known quantity. For each simulation of the 18650 cell, the heat rate $\dot{Q}_{cell}$ was set to 232 mW, while a heat rate of 165 mW was chosen for the 21700 cell. Finally, realistic values for the convective boundaries h_t and h_s were determined.

To determine h_t and h_s , a steady-state optimization study with the thermal model from Section 3.1 was conducted for both cell formats. The objective of this optimization was a congruent temperature increase of T_s , T_{in} , T_{ip} between the model and the experiments displayed in Fig. 4(a). The two air velocities v result in two coefficients at the shell (h_{s1} , h_{s2}) and two at the terminals (h_{t1} , h_{t2}). The arithmetic mean of all the experiments are listed in Table 8 together with their 95% confidence intervals. The results show that higher coefficients for the 18650 cell must be chosen and a positive correlation with the air velocity is observed.

Fig. 7(a) shows the surface temperature T_s of the 18650 cell model at four different through-plane conductivities between 0.4 and 1.4 $\text{W m}^{-1} \text{ K}^{-1}$. For each thermal conductivity value, two simulations were conducted with a heat transfer coefficient at the shell of 15 and 60 $\text{W m}^{-2} \text{ K}^{-1}$. At the terminals, h_{t1} and h_{t2} was set to 40 and 50 $\text{W m}^{-2} \text{ K}^{-1}$, and the heat rate inside the terminals was adjusted to 7% of the total cell heat rate $\dot{Q}_{cell}$. Furthermore, the specific heat capacity of each domain was taken from Table 6 at 300 K. Applying the Bryden method [5] presented in Section 2.2 results in a heat capacity C_{cell} of $(36.0 \pm 1.0) \text{ J K}^{-1}$. This value is in agreement with the volume integral of the specific heat capacity, which is 37.2 J K^{-1} . Therefore, an uncertainty e_{mc} of 5.9% can be specified that arises from the model assumptions of the Bryden method [5], an unknown heat generation $\dot{Q}_t$, and heat convection at the terminal h_t .

Analyzing the simulation data in Fig. 7(a) with Eq. (4) results in four R_{in} values with corresponding $k_{\perp}$. These four value pairs are represented by four red crosses, which are shown in Fig. 7(b) interconnected by a black line. The black line follows a rational function $k_{\perp} = (P_1)/(R_{in} + P_2)$, which contains one asymptote P_2 with a maximum deviation from the red crosses of 0.003 $\text{W m}^{-1} \text{ K}^{-1}$. Therefore, the rational function describes the relation between the lumped thermal resistance R_{in} and the through-plane thermal conductivity $k_{\perp}$ very precisely.

A sensitivity analysis showed a strong linear correlation between P_1 and P_2 with the heat rate ratio $\dot{Q}_t/\dot{Q}_{cell}$ as well as with the difference in heat transfer coefficient at the cell terminals $\Delta h = h_{t2} - h_{t1}$. Eq. (19) is formulated with due regard for this observation, with the five unknown constants p_1 to p_5 . These five constants are determined by fitting Eq. (19) to the results of nine sensitivity simulations with varying Δh of 0, 10, 20 $\text{W m}^{-1} \text{ K}^{-2}$ and $\dot{Q}_t/\dot{Q}_{cell}$ values of 0, 3.5, and 7%. Two of these additional simulations are illustrated in Fig. 7(b) by the red

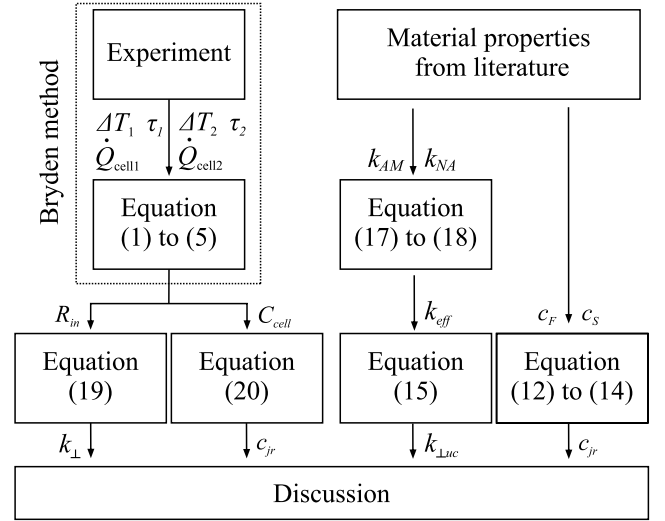


Fig. 8. Block diagram showing the experimental and theoretical approaches for deriving the thermal conductivity and specific heat capacity of cylindrical lithium-ion cells. The left-hand side illustrates the evaluation of the experimental data in Fig. 4. The right-hand side gives an overview of the theoretical determination of $k_{\perp}$ and c_{jr} .

crosses, with the corresponding rational functions in blue and yellow, which display deviations of less than 0.7% between the crosses and the solid lines. Using Eq. (19) and the fitted constants in Table 8, a relation between R_{in} and $k_{\perp}$ can be established without the need for a complex thermal model.

The red crosses in Fig. 7 represent the simulation results of the thermal cell model with a constant thermal conductivity in radial direction $k_{\perp}$, which neglects the spiral geometry. This raises the question of whether the rational Eq. (19) is still valid if the radial conductivity $k_{\perp}$ depends on the radius ($\partial k_{\perp}/\partial r \neq 0$). Gomadam et al. [29] considered the effect of the spiral geometry by increasing $k_{\perp}$ by the quotient $(k_{\parallel} d_{uc}^2)/(4\pi^2 r^2)$. Accordingly, the impact of the spiral depends on the $k_{\parallel}$, the thickness of the unit cell d_{uc} , and the radius r . The increase in $k_{\perp}$ was simulated as an example for the parameter set of the 18650 cell in Fig. 7(a). The result of this simulation is represented by the red circles in Fig. 7(b). By comparing the red circles with the red crosses along the black line, it can be seen that the spiral geometry has a minor influence on the validity of Eq. (19). This agrees with the observation of Gomadam et al. [29] that thermal effects arising from the spiral geometry can be neglected for lithium-ion cells.

In conclusion, the through-plane conductivity $k_{\perp}$ can be evaluated with Eq. (19). The function parameters p_1 to p_5 for both cell formats and Δh_t are listed in Table 8. The heat rate ratio $\dot{Q}_t/\dot{Q}_{cell}$ is calculated with Eq. (1) using the ohmic resistances Z_i in Table 3. The values for R_{in} and C_{cell} are listed in Tables 4 and A.14. Most experimental studies in the field of lithium-ion cells do not evaluate the specific heat of the electrode stack or jelly roll, which renders comparisons with material measurements difficult. Instead, the specific heat capacity of the total cell $c_{cell} = C_{cell}/m_{cell}$ is evaluated [13] [4]. To enable comparison with the heat capacity estimations in Section 3.2, the heat capacity of the jelly roll c_{jr} is defined by Eq. (14), which compensates for the non-active cell components (m_{na} , c_{na}) listed in Table 6.

$$k_{\perp} = \frac{p_1 + p_2 \Delta h_t}{R_{in} + p_3 + p_4 \frac{\dot{Q}_t}{\dot{Q}_{cell}} + p_5 \Delta h_t} \quad (19)$$

$$c_{jr} = \frac{\frac{C_{cell}}{m_{cell}} - \frac{m_{na}}{m_{cell}} c_{na}}{1 - \frac{m_{na}}{m_{cell}}} \quad (20)$$

Table 8

Parameters in Eq. (19) for calculating $k_{\perp}$ based on the lumped thermal resistance R_{in} .

	18650	21700
p_1 / m^{-1}	0.5660	0.5120
$p_2 / \text{K m W}^{-1}$	-0.0017	-0.0022
$p_3 / \text{K W}^{-1}$	0.0616	0.0552
$p_4 / \text{K W}^{-1}$	-0.4408	-0.4448
$p_5 / \text{K}^2 \text{m}^2 \text{W}^{-2}$	0.0018	0.0024
$h_{t1} / \text{W m}^{-2} \text{K}^{-1}$	42.0 ± 1.4	34.7 ± 1.1
$h_{t2} / \text{W m}^{-2} \text{K}^{-1}$	48.5 ± 2.2	35.4 ± 1.7
$h_{s1} / \text{W m}^{-2} \text{K}^{-1}$	14.3 ± 0.4	11.3 ± 0.3
$h_{s2} / \text{W m}^{-2} \text{K}^{-1}$	61.4 ± 0.3	43.8 ± 0.8

Table 9

Systematic measurement and model uncertainties of the Bryden method [5] and model simplifications from Eqs. (19) and (20).

Parameter	Uncertainty
R_{in}	$e_R = \pm 5.19\%$
$k_{\perp}$	$e_k = \pm 6.16\%$
C_{cell}	$e_C = \pm 1.62\%$
$c_{cell} = C_{cell}/m_{cell}$	$e_c = \pm 4.59\%$
c_{jr}	$e_{cjr} = \pm 4.71\%$

Eq. (19) introduces additional uncertainties for $k_{\perp}$, which have to be added to the measurement uncertainty detailed in Section 2.3. The largest uncertainty in Eq. (19) originates from the parameter Δh . For example, an uncertainty in Δh of $3.6 \text{ W m}^{-1} \text{K}^{-2}$ can translate to an uncertainty in $k_{\perp}$ of 3.31%. This uncertainty and the measurement uncertainty e_R of 5.19% by the root sum square result in a total uncertainty for $k_{\perp}$ of 6.16%. Both uncertainties for R_{in} and $k_{\perp}$ are presented in Table 9.

To estimate the uncertainty e_c of c_{cell} , the root sum square of e_{mc} and e_C are used, resulting in $e_c = \pm 4.59\%$. This uncertainty can be compared with differential scanning calorimeters, which offer an uncertainty between 1.5 and 10% [30]. With lithium-ion cells, the derived value for e_c is close to the uncertainty of 3.82% of the technique developed by Sheng et al. [13], but worse than the uncertainty of 0.42% of the calorimeter used by Bazinski et al. [31].

For c_{jr} , an uncertainty for c_{na} of 10% is assumed, which is the upper bound of the uncertainty range of the differential scanning calorimetry [30]. Taking the smallest value for C_{cell} in Table 4 of 42.2 J K^{-1} and a value of $570.1 \text{ J kg}^{-1} \text{K}^{-1}$ from Table 6 for c_{na} , the specific heat capacity of the jelly roll c_{jr} calculates to $971 \text{ J kg}^{-1} \text{K}^{-1}$. Reducing c_{na} by 10% in this calculation increases c_{jr} ($981 \text{ J kg}^{-1} \text{K}^{-1}$), which constitutes an additional uncertainty of 1.06% for c_{jr} . Finally, the root sum square of e_c and 1.06% computes to $e_{cjr} = 4.71\%$ for the specific heat capacity of the jelly roll. It should be noted that a full validation of the uncertainties in Table 9 is beyond the scope of this paper. While the authors have a high level of confidence in the calculated uncertainties, the proposed measurement methods are so far conceptual and require further research in order to validate their robustness.

4. Results and discussion

This section discusses the heat capacity and thermal conductivity of a 18650 and a 21700 lithium-ion cell. An experimental and a theoretical approach were chosen to determine both thermal properties. The experimental approach is based on the measurements in Section 2, which are evaluated using the theory presented in Section 3.4. The process is illustrated on the left-hand side of Fig. 8. The experiments are conducted and evaluated according to the Bryden method [5]. Afterwards, R_{in} is transformed to $k_{\perp}$, which represents the extension of this work. Furthermore, the heat capacity of the cell C_{cell} is converted to the specific heat capacity of the jelly roll c_{jr} using Eq. (20).

Table 10

Linear regression parameters for the measured specific heat capacity of the jelly roll $c_{jr} = c_0 + c_1(T - 300 \text{ K})$ in Fig. 9 with the respective temperature dependencies c_1/c_0 . The given confidence intervals describe the uncertainty of the regression parameters. Systematic uncertainties of c_{jr} can be found in Table 9.

$\frac{U_{ocv}}{V}$	c_0 $\text{J kg}^{-1} \text{K}^{-1}$	c_1 $\text{J kg}^{-1} \text{K}^{-2}$	c_1/c_0 $\% \text{K}^{-1}$
18650 cell format			
3.42	1003.4 ± 1.2	1.9 ± 0.1	0.2 ± 0.01
3.64	1003.6 ± 2.6	2.2 ± 0.2	0.2 ± 0.06
3.93	1002.5 ± 1.2	2.1 ± 0.1	0.2 ± 0.03
21700 cell format			
3.42	908.0 ± 2.1	1.6 ± 0.2	0.2 ± 0.02
3.64	912.0 ± 2.7	1.6 ± 0.2	0.2 ± 0.04
3.93	910.5 ± 1.3	1.9 ± 0.1	0.2 ± 0.05

Table 11

Linear regression parameters for the measured through-plane thermal conductivity $k_{\perp} = k_0 + k_1(T - 300 \text{ K})$ in Fig. 10 with the respective temperature dependencies k_1/k_0 . The given confidence intervals describe the uncertainty of the regression parameters. Systematic uncertainties of $k_{\perp}$ can be found in Table 9.

$\frac{U_{ocv}}{V}$	k_0 $\text{W m}^{-1} \text{K}^{-1}$	k_1 $\text{mW m}^{-1} \text{K}^{-2}$	k_1/k_0 $\% \text{K}^{-1}$
18650 cell format			
3.42	1.225 ± 0.033	-7.4 ± 2.8	-0.6 ± 0.25
3.64	1.288 ± 0.041	-5.1 ± 3.7	-0.4 ± 0.30
3.93	1.298 ± 0.032	-6.1 ± 2.7	-0.5 ± 0.22
21700 cell format			
3.42	1.245 ± 0.049	-6.5 ± 4.1	-0.5 ± 0.35
3.64	1.357 ± 0.072	-13.4 ± 6.5	-1.0 ± 0.53
3.93	1.348 ± 0.047	-5.9 ± 3.9	-0.4 ± 0.30

Thereafter, the experimental values for $k_{\perp}$ and c_{jr} can be compared with the theoretical approach, which is shown on the right-hand side of Fig. 8. For the thermal conductivity $k_{\perp,uc}$, the effective conductivity k_{eff} of each porous cell layer is calculated based on the theory of Bruggeman given in Section 3.3. For the specific heat capacity of the jelly roll c_{jr} , the heat capacities of the fluid c_F and solid c_S phase are considered in Section 3.2. The thermal conductivities $k_{\perp}$ and $k_{\perp,uc}$ are treated differently in this study, as $k_{\perp}$ can contain thermal contact resistances as well as effects of the spiral geometry of the jelly roll.

4.1. Specific heat capacity

Fig. 9 shows the specific heat capacity of the jelly roll c_{jr} for three different temperatures and open circuit voltages U_{ocv} . Each measurement is represented by a dot and the solid lines display the linear regressions for each open circuit voltage. The constant c_0 and the slope c_1 of the linear regression together with the 95% confidence interval are listed in Table 10. Dividing the slope of the regression c_1 by the absolute value at 300 K c_0 gives the temperature dependency of the specific heat capacity.

The range of c_{jr} lies within 850 and $1050 \text{ J kg}^{-1} \text{K}^{-1}$ for both cell formats. This agrees well with the range reported by Loges et al. [24] for wet NMC unit cells of 900 to $1100 \text{ J kg}^{-1} \text{K}^{-1}$. In comparison with the theoretical estimations for c_0 given in Table 6, the measurement results at 300 K are 20% higher for the 18650 cell and 4% higher for the 21700 cell. The divergence between the measurement and the calculation could result from uncertainties in both techniques. According to Tables 9 and A.19, uncertainties justify a deviation of 12.91%, which explains the deviation of 4% for the 21700 cell but not the 20% of the 18650 cell. Further uncertainties for the theoretical c_{jr} can arise from the assumption that the pore and electrolyte volumes are equal. In reality, the volume of the electrolyte can exceed the volume of the pores [32]. For the 18650 cell, an additional electrolyte could explain the deviation of 20%, although this would worsen the match between the measured and calculated cell weights in Table 6.

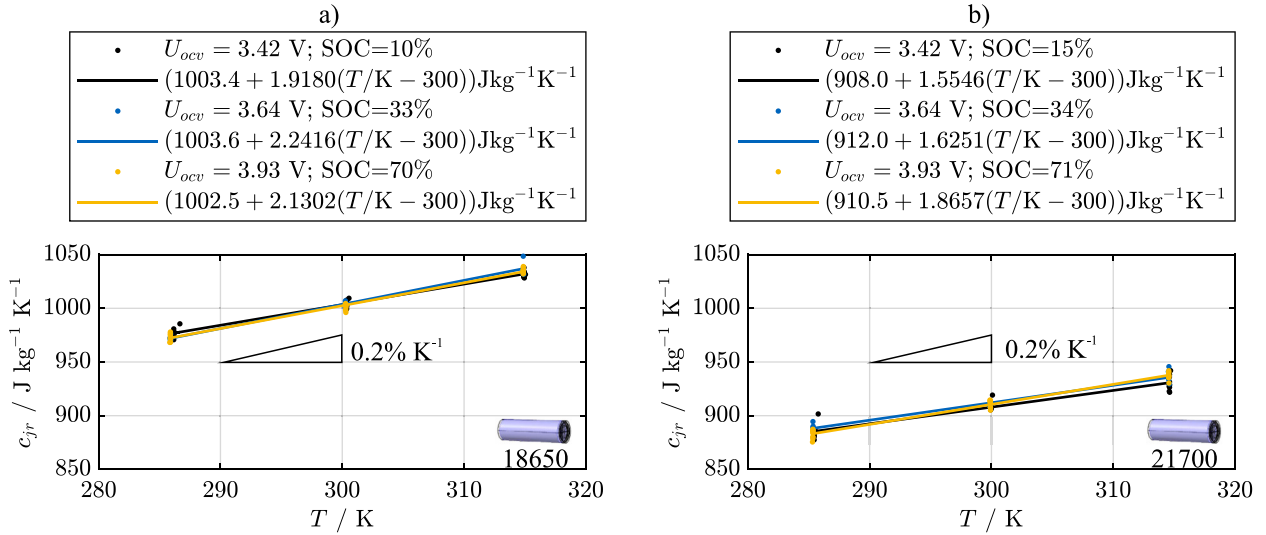


Fig. 9. Measured specific heat capacity of the jelly roll according to Eq. (20) with linear regressions at different open circuit voltages U_{ocv} . (a) 18650 cell. (b) 21700 cell.

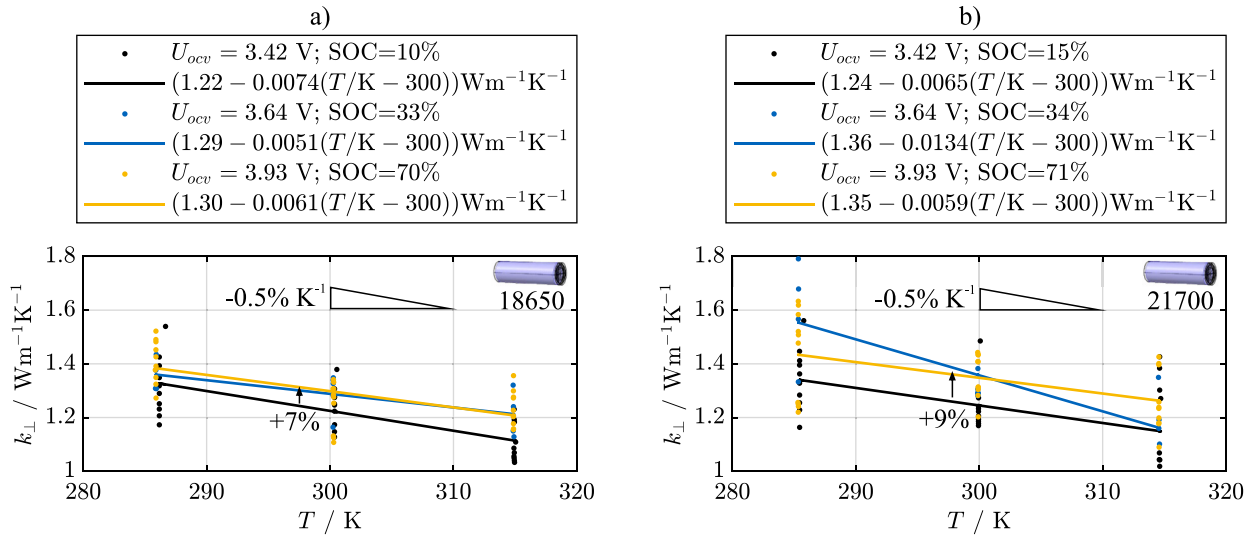


Fig. 10. Measured thermal conductivity of the jelly roll in through-plane direction according to Eq. (19) with linear regressions at different open circuit voltages U_{ocv} . (a) 18650 cell. (b) 21700 cell.

The heat capacities of both cell formats increase equally by $0.2\% \text{ K}^{-1}$ with increasing temperature. This increase is independent from the SOC and agrees with the theoretical estimation in Table 6. No significant dependency on the SOC is observed either for the 18650 or for the 21700 format. Small changes in c_0 in Table 10 are within the given confidence interval of the linear regressions. The independence of c_{jr} from the SOC agrees with the observations of Bazinski et al. [31].

4.2. Through-plane conductivity

Fig. 10 shows the results for $k_{\perp}$ at varying SOC and temperatures. The quotient $\dot{Q}_t/\dot{Q}_{cell}$ in Eq. (19) lies within 4% to 9% for the 18650 cell. This change is mainly caused by changing heat losses inside the jelly roll at different ambient temperatures T_{amb} , while the heat losses inside the terminals remain constant. For the 21700 cell, $\dot{Q}_t/\dot{Q}_{cell}$ varies between 6% and 10%. The results for $k_{\perp}$ in Fig. 10 are presented in the same manner as for c_{jr} . This also applies to Table 11, which lists the constant k_0 and the slope k_1 of the linear regressions in Fig. 10, together

with the 95% confidence interval and the temperature dependency k_1/k_0 .

The thermal conductivity at 300 K and $U_{ocv} = 3.64 \text{ V}$ amounts to 1.29 and $1.36 \text{ W m}^{-1} \text{K}^{-1}$ for the 18650 and 21700 cell, respectively. This constitutes a deviation of 15% and 26% compared to the theoretical values for $k_{\perp uc}$ derived with Eq. (15). Deviations of this magnitude are common in the literature. For example, Sheng et al. [13] observed a deviation between the measured and calculated through-plane conductivity of 56%. According to Tables 9 and A.19, the deviation between $k_{\perp}$ and $k_{\perp uc}$ of 15% (18650) and 26% (21700) in this paper can be explained by the calculated uncertainty of $\pm 27.56\%$ ($\pm(6.16 + 21.4)\%$). One explanation for the observed deviation could be that the coating porosities of the 18650 cell are below 20%. According to Eq. (18), a low porosity shifts the effective conductivity towards the conductivity of the coating composite, which is usually higher than the conductivity of the electrolyte. The exact porosity of the 18650 cell is considered in the calculations illustrated in Fig. 6(c). By comparing k_0 in Table 11 at $U_{ocv} = 3.42 \text{ V}$ with $k_{\perp uc}$ from A3C3 at 300 K, it can be seen that there is a deviation of 11% from the measurement. This overestimation of A3C3

might be explained by the thermal contact resistances between the cell layers. It should also be mentioned that the measurement results $k_{\perp}$ can be higher than the calculated $k_{\perp,uc}$ if the spiral geometry of the jelly roll causes a significant in-plane heat flow inside the current collectors [20].

A lower thermal conductivity $k_{\perp,uc}$ was computed in Section 3.2 for the 21700 format, which is contradicted by the measurement results in Fig. 10. For example, $k_{\perp}$ is 4% higher for the 21700 format at an open circuit voltage of 3.93 V. This can be explained by thermal contact resistances between the jelly roll and the can. For example, Gaitonde et al. [19] measured a thermal conductance of $650 \text{ W m}^{-2} \text{ K}^{-1}$ for this interface. Due to the higher radius of the 21700 cell, the impact of this heat barrier is smaller than for the 18650 format.

The comparison of k_0 in Table 11 shows increasing $k_{\perp}$ with an increasing open circuit voltage of 7% and 9% for the 18650 and 21700 cell, respectively. This observation is in good accordance with the estimated 12% increase in $k_{\perp,uc}$ found in Fig. 6(c). Consequently, it is reasonable to conclude that the changing thermal conductivity of the pristine graphite particle is responsible for this increase. Moreover, a thickness change in the electrode coatings induced by changes in SOC can cause an increase in the $k_{\perp}$. With increasing SOC, the graphite coating thickness increases more strongly than the NMC coating thickness decreases [33], which enhances the weight of the k_{anode} in Eq. (15). As the thermal conductivity of the anode is greater than the conductivity of the cathode, the $k_{\perp,uc}$ will increase with SOC. Furthermore, the thickness change of the whole unit cell increases the mechanical stress inside the jelly roll [34], which might reduce the thermal contact resistance between the cell layers.

The temperature dependency k_1/k_0 in Fig. 10 lies within -0.4 and $-0.6 \text{ \% K}^{-1}$ for both cell formats. The high value of $-1.0 \text{ \% K}^{-1}$ for the 21700 format in Table 11 can be treated as an outlier. The derived values for k_1/k_0 are in accordance with the theoretical range in Fig. 6(c), which is between -1.7 and $-0.2 \text{ \% K}^{-1}$. In the literature, there is no agreement in regard to the temperature dependency of the through-plane conductivity $k_{\perp}$. For example, Sheng et al. [13] [4] and Liebig et al. [35] observed an increase in $k_{\perp}$ with increasing temperature. On the other hand, a decrease in conductivity with increasing temperature was measured by Werner et al. [11], Steinhardt et al. [3] and Murashko et al. [36].

5. Conclusion

This paper introduces a simple and precise method for measuring thermal conductivity, which also characterizes thermal heat capacity. The evaluation of both thermophysical properties requires only standard laboratory equipment, needs no special cell preparation, and saves a complex evaluation of the measurement data. To achieve a state-of-the-art level of uncertainty of 6.16% for thermal conductivity, the measurement settings have to be chosen carefully, which is described in this paper. Despite the confidence in the calculated uncertainty of 6.16%, further experiments are necessary to fully validate the robustness of the conceptual measurement method for the thermal conductivity. The method is demonstrated for one cylindrical 18650 and one 21700 cell at three different ambient temperatures and states of charge. For both cell formats, the state of charge has no dominant effect on the heat capacity. Only the cell temperature was found to affect the specific heat capacity of the jelly roll by $0.2 \text{ \% K}^{-1}$. A rational function was used to remove the need for a complex finite element model to evaluate the measurement data. The through-plane conductivity of the electrode stack was evaluated and was found to decrease with increasing temperature by $-0.5 \text{ \% K}^{-1}$. Furthermore, an increasing thermal conductivity of 7% to 9% was observed with an increasing state of charge of between 10% and 71%. This increase could be due to the changing conductivity of the active materials or changing thermal contact resistances. Future work will concentrate on adapting the presented method for pouch and prismatic cell formats as well as on the effects of state of health on thermal conductivity. In

addition, more research on the effects of thermal contact resistance and the spiral geometry of the jelly roll will be necessary to quantify their impact on thermal conductivity in through-plane direction. Finally, more repeated tests on different cells of the same type are planned to further increase the degree of trust in the precision of the proposed thermal characterization methods.

CRedit authorship contribution statement

Marco Steinhardt: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft. **Elisabeth Irene Gillich:** Conceptualization, Methodology, Writing – review & editing, Funding acquisition. **Alexander Rheinfeld:** Methodology, Writing – review & editing, Funding acquisition. **Ludwig Kraft:** Methodology, Writing – review & editing. **Markus Spielbauer:** Resources, Writing – review & editing. **Oliver Bohlen:** Resources, Writing – review & editing. **Andreas Jossen:** Conceptualization, Supervision, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Nomenclature can be found in Table A.12, Table A.13

Table A.12

Nomenclature: Greek symbols.

Symbols	
β	Bruggemann coefficient
$\bar{\rho}$	Bulk density of porous cell layer in kg m^{-3}
ρ_{AM}	Density of active material in kg m^{-3}
ρ_{BCB}	Density of binder and carbon black in kg m^{-3}
ρ_F	Density of fluid phase in kg m^{-3}
ρ_S	Density of solid phase in kg m^{-3}
$\bar{\rho}_l$	Bulk density of cell layer in kg m^{-3}
ρ_{jr}	Density of jelly roll in kg m^{-3}
ε_{AM}	Volume fraction of active material
ε_{BCB}	Volume fraction of binder and carbon black
ε_S	Volume fraction of solid phase
ε_F	Volume fraction of fluid phase or porosity
φ	Phase angle of voltage with respect to current in rad
τ_1	Time constant at v_1 in s
τ_2	Time constant at v_2 in s
$\Delta\tau$	Difference between τ_1 and τ_2 in s

Table A.13

Nomenclature.

Symbols	
C_{cell}	Lumped heat capacity of the cell JK^{-1}
$\bar{c}$	Bulk specific heat capacity of porous layer in $\text{Jkg}^{-1}\text{K}^{-1}$
c_{jr}	Specific heat capacity of the jelly roll in $\text{Jkg}^{-1}\text{K}^{-1}$
c_{na}	Specific heat capacity of non-active cell parts in $\text{Jkg}^{-1}\text{K}^{-1}$
c_{ao}	Specific heat capacity of anode overhang in $\text{Jkg}^{-1}\text{K}^{-1}$
$d_{jr,s}$	Diameter of the jelly roll gap in m
$d_{ll,i}$	Total length unit cell layer in m
$d_{lt,i}$	Thickness unit cell layer in m
d_{uc}	Total thickness of the unit cell in m
$e_{\Delta\tau}$	Measurement uncertainty in $\Delta\tau$
e_{τ_1/τ_2}	Measurement uncertainty in the ratio of τ_1 and τ_2
e_{mc}	Modeling uncertainty in C_{cell}
e_C	Measurement uncertainty in C_{cell}
e_R	Measurement uncertainty in R_{in}
e_{Rout1}	Measurement uncertainty in R_{out1}
e_{Rout2}	Measurement uncertainty in R_{out2}
e_I	Measurement uncertainty in $\hat{I}_{cell}$
e_U	Measurement uncertainty in $\hat{U}_{cell}$
e_k	Uncertainty in $k_{\perp}$
e_c	Uncertainty in c_{cell}
e_{jr}	Uncertainty in c_{jr}
f	Uncertainty gain factor
h_{s1}	Heat transfer coefficient at the shell of the cell at v_1 in $\text{Wm}^{-2}\text{K}^{-1}$
h_{s2}	Heat transfer coefficient at the shell of the cell at v_2 in $\text{Wm}^{-2}\text{K}^{-1}$
h_{t1}	Heat transfer coefficient at the terminal of the cell at v_1 in $\text{Wm}^{-2}\text{K}^{-1}$
h_{t2}	Heat transfer coefficient at the terminal of the cell at v_2 in $\text{Wm}^{-2}\text{K}^{-1}$
Δh	Difference between h_{t2} and h_{t1} in $\text{Wm}^{-2}\text{K}^{-1}$
I_{cell}	Amplitude of sinusoidal cell current in A
I_{cell}	Electrical cell current A
k_{Luc}	Thermal conductivity (through-plane) of the unit cell in $\text{Wm}^{-1}\text{K}^{-1}$
$k_{\perp}$	Thermal conductivity (through-plane) of the jelly roll in $\text{Wm}^{-1}\text{K}^{-1}$
$k_{ }$	Thermal conductivity (in-plane) of the jelly roll and unit cell in $\text{Wm}^{-1}\text{K}^{-1}$
k_{AM}	Thermal conductivity of the active material in $\text{Wm}^{-1}\text{K}^{-1}$
k_{NA}	Thermal conductivity of the non-active material in $\text{Wm}^{-1}\text{K}^{-1}$
$k_{pristine}$	Thermal conductivity of pristine material in $\text{Wm}^{-1}\text{K}^{-1}$
k_{BCB}	Thermal conductivity of the binder and carbon black in $\text{Wm}^{-1}\text{K}^{-1}$
k_{dry}	Thermal conductivity of porous material without electrolyte in $\text{Wm}^{-1}\text{K}^{-1}$
k_{wet}	Thermal conductivity of porous material with electrolyte in $\text{Wm}^{-1}\text{K}^{-1}$
k_{eff}	Thermal conductivity derived by the Bruggemann model in $\text{Wm}^{-1}\text{K}^{-1}$
k_F	Thermal conductivity of fluid phase in $\text{Wm}^{-1}\text{K}^{-1}$
k_i	Thermal conductivity of cell layers in $\text{Wm}^{-1}\text{K}^{-1}$
k_r	Thermal conductivity in radial direction in $\text{Wm}^{-1}\text{K}^{-1}$
k_z	Thermal conductivity in axial direction in $\text{Wm}^{-1}\text{K}^{-1}$
m_{cell}	Mass of the cell in kg
m_i	Mass of the cell layers in kg
m_{jr}	Mass of jelly roll in kg
m_{ao}	Mass of anode overhang in kg
m_{na}	Mass of the non jelly roll domains in kg
$\dot{Q}_{cell1}, \dot{Q}_{cell2}$	Heat rate inside the cell at v_1 and v_2 in W
$\dot{Q}_{jr}$	Heat rate inside the jelly roll in W
$\dot{Q}_t$	Heat rate inside the terminals in W
$\dot{q}$	Volume specific heat source in Wm^{-3}
R_{in}	Thermal resistance inside the cell in KW^{-1}
R_{out1}, R_{out2}	Thermal resistance outside the cell at v_1 and v_2 in KW^{-1}
t	Time in s
T_{amb}	Ambient temperature of the cell in K
T_{in}	Temperature at negative terminal of the cell in K
T_{tp}	Temperature at positive terminal of the cell in K
T_s	Temperature at the shell of the cell in K
$\Delta T_{s1}, \Delta T_{s2}$	Temperature increase at the shell of the cell at v_1 and v_2 in K
U_{cell}	Cell voltage in V
U_{ocv}	Open circuit voltage in V
$\hat{U}_{cell}$	Amplitude of sinusoidal cell voltage in V
v_1, v_2	Lower and upper air velocity in ms^{-1}
V_{jr}	Volume of jelly roll in m^3
V_{ao}	Volume of anode overhang in m^3
Z_i	Internal electrical impedance in Ω
Z_e	External electrical impedance in Ω

Table A.14

Measurement results for the lumped parameters heat capacity C_{cell} and internal resistance R_{in} for the 21700 cell at three different ambient temperatures and SOCs. For each result, the arithmetic mean and the 95% interval is given.

$\frac{U_{ocv}}{V}$	$\frac{T_{amb}}{K}$	$\frac{R_{in}}{K W^{-1}}$	$\frac{C_{cell}}{J K^{-1}}$	$\frac{R_{out1}}{K W^{-1}}$	$\frac{R_{out2}}{K W^{-1}}$	$\frac{\tau_1}{s}$	$\frac{\tau_2}{s}$
3.42	283	0.358 ± 0.02	56.8 ± 0.27	13.5 ± 0.05	4.7 ± 0.00	784.8 ± 0.84	286.4 ± 0.70
	298	0.396 ± 0.02	58.3 ± 0.16	14.0 ± 0.04	4.7 ± 0.01	839.6 ± 1.05	299.7 ± 0.53
	313	0.437 ± 0.05	59.7 ± 0.27	14.0 ± 0.03	4.8 ± 0.01	864.8 ± 1.60	312.5 ± 0.95
3.63	283	0.302 ± 0.07	57.1 ± 0.31	13.4 ± 0.12	4.7 ± 0.04	782.6 ± 6.26	283.8 ± 1.19
	298	0.375 ± 0.01	58.4 ± 0.05	14.0 ± 0.07	4.7 ± 0.02	836.9 ± 3.57	296.8 ± 1.21
	313	0.415 ± 0.06	60.1 ± 0.66	14.0 ± 0.04	4.7 ± 0.05	866.2 ± 8.22	310.4 ± 1.86
3.93	283	0.339 ± 0.03	56.7 ± 0.16	13.3 ± 0.03	4.6 ± 0.00	775.7 ± 0.89	281.2 ± 1.28
	298	0.363 ± 0.02	58.5 ± 0.13	14.1 ± 0.05	4.8 ± 0.02	844.0 ± 2.22	300.9 ± 1.41
	313	0.395 ± 0.03	60.1 ± 0.13	14.0 ± 0.03	4.8 ± 0.01	868.1 ± 2.03	312.4 ± 1.15

Table A.15

Material densities and temperature-dependent heat capacities $c/(J\text{kg}^{-1}\text{K}^{-1}) = c_{m3}T^{-3} + c_{m2}T^{-2} + c_0 + c_1T + c_2T^2 + c_3T^3$.

	$\frac{\rho}{\text{g cm}^{-3}}$	$\frac{c_{m3}}{10^{-8}}$	$\frac{c_{m2}}{10^{-6}}$	$\frac{c_0}{1}$	$\frac{c_1}{1}$	$\frac{c_2}{10^{-3}}$	$\frac{c_3}{10^{-4}}$
Graphite [21,37]	2.30	0	0	$684.12.996$	$^{\circ}\text{C}^{-1}$	0	0
NMC [21,37]	4.70	0	0	$757.11.978$	$^{\circ}\text{C}^{-1}$	0	0
PP Separator [21,38]	0.91	0	0	$1148.7.938$	$^{\circ}\text{C}^{-1}$	15.24	$^{\circ}\text{C}^{-2}$
Cu [38,39]	8.96	0	0	356	0.0988	K^{-1}	0
Al [38,40]	2.70	2.84	K^3	-6.64	K^2	785	0.906
Steel 304 [41,42] ^a	7.90	0	0	$182.11.6185$	K^{-1}	-2.02	K^{-2}
Electrolyte [21]	1.24	0	0	$1610.1.398$	$^{\circ}\text{C}^{-1}$	4.457	$^{\circ}\text{C}^{-2}$

^aRecommended curve between 200 and 300 K.

Table A.16

Temperature-dependent thermal conductivity of dry and pristine materials used in lithium-ion cells.

Material	278 K	288 K	298 K	308 K	318 K
$k/\text{W m}^{-1} \text{ K}^{-1}$					
Dry and porous:					
A1: Graphite in CO ₂ [21]	2.46	2.34	2.08	1.87	1.79
C1: NMC in CO ₂ [21]	0.24	0.20	0.17	0.15	0.14
C2: NMC delithiated in argon [28]	0.344	0.341	0.338	0.335	0.333
C3: NMC lithiated in argon [28]	0.441	0.437	0.433	0.429	0.426
Pristine and non-porous:					
A2: Pyrolytic graphite $\perp$ [26]	10.4	9.99	9.58	9.26	8.96
A3: 33% of A2 [26,27]	3.42	3.30	3.16	3.06	2.96
PVDF [43]	0.282	0.273	0.263	0.253	0.244
LP50 [11]	0.180	0.177	0.175	0.172	0.170
CO ₂ [44]	0.0151	0.0158	0.0166	0.0175	0.0183
Argon [44]	0.0166	0.0171	0.0176	0.0181	0.0186
Cu [23]	400.4	399.3	398.2	397.3	396.5
Al [23]	236.2	236.6	236.9	237.5	238.1

Table A.17

Structural parameters of wet 21700 unit cell layers and their temperature-dependent heat capacity $\bar{c}/(\text{Jkg}^{-1}\text{K}^{-1}) = c_0 + c_1(T/K - 300)$.

	ϵ_F^a	ϵ_{BCB}^a	$\frac{d_{ll,i}}{\mu\text{m}}$ [9]	$\frac{d_{ll,i}}{\text{m}}$ [9] ^b	$\frac{\bar{\rho}_i}{\text{gcm}^{-3}}$ ^c	$\frac{m_i}{\text{g}}$ ^d	c_0^e	c_1^e
Anode	0.300	0.090	96.5	1.666	1.95	19.9	934.8	2.733
Cathode	0.300	0.084	76.2	1.604	3.43	26.6	898.1	1.727
Separator	0.450	0.000	18.0	1.810	1.06	2.2	1515.5	4.573
Cu-collector	0.000	0.000	12.0	0.905	8.96	6.2	385.6	0.099
Al-collector	0.000	0.000	13.6	0.811	2.70	1.9	892.4	0.635

^aValues originate from the 18650 cell except for ϵ_F of the anode and cathode, which are adjusted to match the measured and calculated cell weights in Table 6.

^bFor double-coated electrodes, the length is accumulated.

^cBulk densities of porous layer $\bar{\rho}_i$ are calculated with Eq. (12) and the material densities in Table A.15.

^dThe total layer mass m_i is derived from the product of the layer thickness $d_{ll,i}$ and length $d_{ll,i}$, the bulk density $\bar{\rho}_i$ and the active jelly roll height of 63.5 mm [9].

^eCalculated with Eq. (14).

Table A.18

Equations and boundary condition of the thermal model in Section 3.1 and Fig. 5.

Heat equation	
$\rho p(r, z) c(r, z) \frac{\partial T(r, z, t)}{\partial t} = \vec{\nabla} \cdot \left(\begin{pmatrix} rk_r(r, z) & 0 \\ 0 & rk_z(r, z) \end{pmatrix} \vec{\nabla} T(r, z, t) \right) + r \dot{q}(r, z, t)$	
Initial temperature	
$T(r, z, t = 0 \text{ s}) = T_{amb}$	
Heat source	
Domain 1–5	$\dot{q}(r, z, t) = \dot{Q}_i / (\sum_{n=1}^5 V_n)$
Domain 6	$\dot{q}(r, z, t) = \dot{Q}_{jr} / (V_6)$
Boundary conditions	
Terminals	$\vec{n} \cdot \begin{pmatrix} rk_r(r, z) & 0 \\ 0 & rk_z(r, z) \end{pmatrix} \vec{\nabla} T(r, z, t) = rh_i(T_{amb} - T(r, z, t))$
Shell	$\vec{n} \cdot \begin{pmatrix} rk_r(r, z) & 0 \\ 0 & rk_z(r, z) \end{pmatrix} \vec{\nabla} T(r, z, t) = rh_s(T_{amb} - T(r, z, t))$

Table A.19

Uncertainty analysis for the specific heat capacity of the jelly roll c_{jr} in Eq. (14) and for the through-plane conductivity of the unit cell $k_{\perp}$ defined by Eq. (15). Every estimation e_i is based on the 18650 cell with $k_{\perp} = 1.122 \text{ W m}^{-1} \text{ K}^{-1}$ and $c_{jr} = 834 \text{ J kg}^{-1} \text{ K}^{-1}$.

Argument uncertainty	$e_i = \frac{\Delta k_{\perp}}{k_{\perp}}$	$e_i = \frac{\Delta c_{jr}}{c_{jr}}$
163% · k_{anode}^a	17.1%	–
126% · $k_{cathode}^a$	11.5%	–
184% · $k_{separator}^a$	5.9%	–
101% · k_{Cu}^a	0.0%	–
101% · k_{Al}^a	0.0%	–
101% · $d_{it,anode}^b$	0.1%	0.0%
102% · $d_{it,cathode}^b$	0.2%	0.0%
108% · $d_{it,separator}^b$	0.5%	0.2%
109% · $d_{it,Cu}^b$	0.3%	0.5%
106% · $d_{it,Al}^b$	0.3%	0.0%
115% · $\bar{c}_{anode}^{c,d,e}$	–	5.5%
112% · $\bar{c}_{cathode}^{c,d,f}$	–	6.0%
112% · $\bar{c}_{separator}^{c,d,g}$	–	0.6%
101% · $\bar{c}_{Cu}^h$	–	0.1%
101% · $\bar{c}_{Al}^h$	–	0.1%
$\sqrt{\sum e_i^2}$	21.4%	8.2%

^aUncertainties are taken from [3].^bUncertainties are based on a precision of 1 μm for common micrometer screws.^cConsiders an uncertainty of 2% [28] for ϵ_F .^dConsiders an uncertainty of 10.7% for c_F , which is determined by the deviation between [24] and [45].^eConsiders an uncertainty of 19.5% for c_S [24,46].^fConsiders an uncertainty of 13.7% for c_S [24,46].^gConsiders an uncertainty of 25.3% for c_S [24,47].^hConsiders an uncertainty of 1.5% for c_S , which is the lower uncertainty bound for differential scanning calorimeters [30].**Table A.20**

Parameters of the thermal model in Section 3.1 and Fig. 5.

Domain	$\frac{\rho}{\text{g cm}^{-3}}$	$\frac{c}{\text{J kg}^{-1} \text{ K}^{-1}}$	$\frac{k_r}{\text{W m}^{-1} \text{ K}^{-1}}$	$\frac{k_z}{\text{W m}^{-1} \text{ K}^{-1}}$
1, 7	Table A.15	Table A.15	Table A.16	$k_r = k_z$
2, 3	Table A.15	Table A.15	Table A.16	$k_r = k_z$
4	Table A.15	Table A.15	Table A.16	$k_r = k_z$
5	$m_{ao}/V_{ao}^{a,b}$	c_{ao}^a	$0.041^{c,e}; 0.042^{d,e}$	$12.978^{c,e}; 12.485^{d,e}$
6	$m_{jr}/V_{jr}^{a,b}$	c_{jr}^a	$k_{\perp}$	$k_{\parallel}$

^aTaken from Table 6.^bTaken from Fig. 5.^cApplied for the 18650 cell.^dApplied for the 21700 cell.^eAnalogous calculation as for $k_{\perp}$ and $k_{\parallel}$ with Eqs. (15) and (16) but with the Al collector and cathode layer replaced by the thermal conductivity of argon in Table A.16.

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